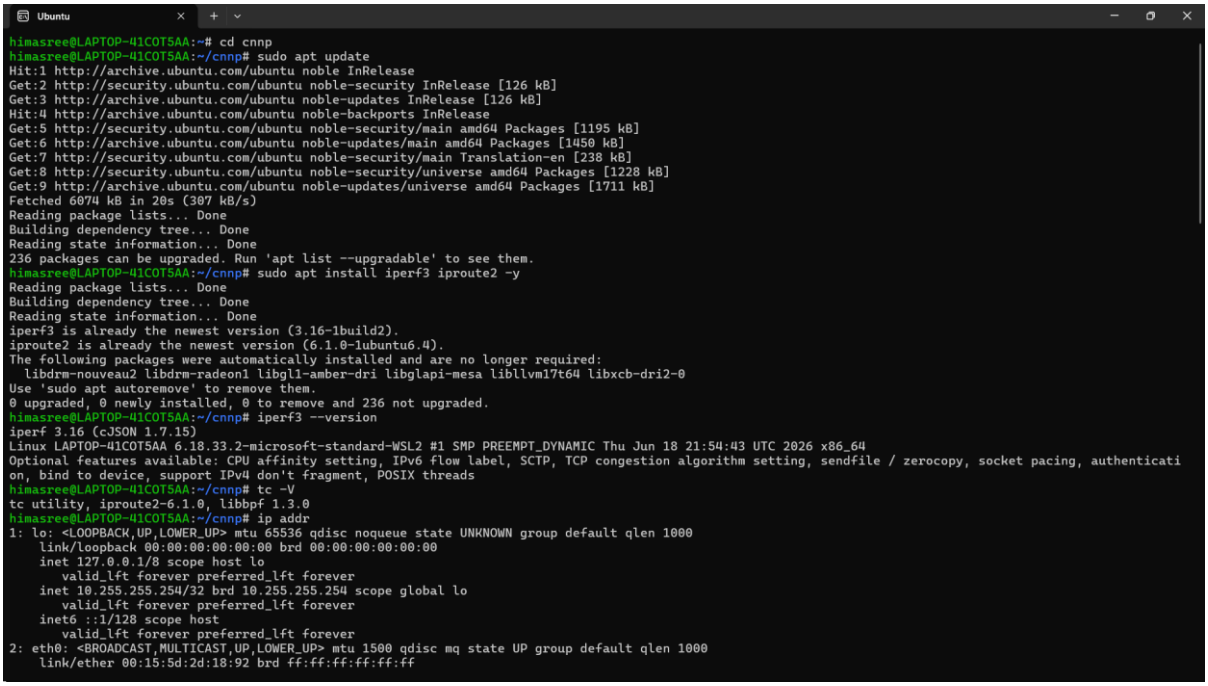
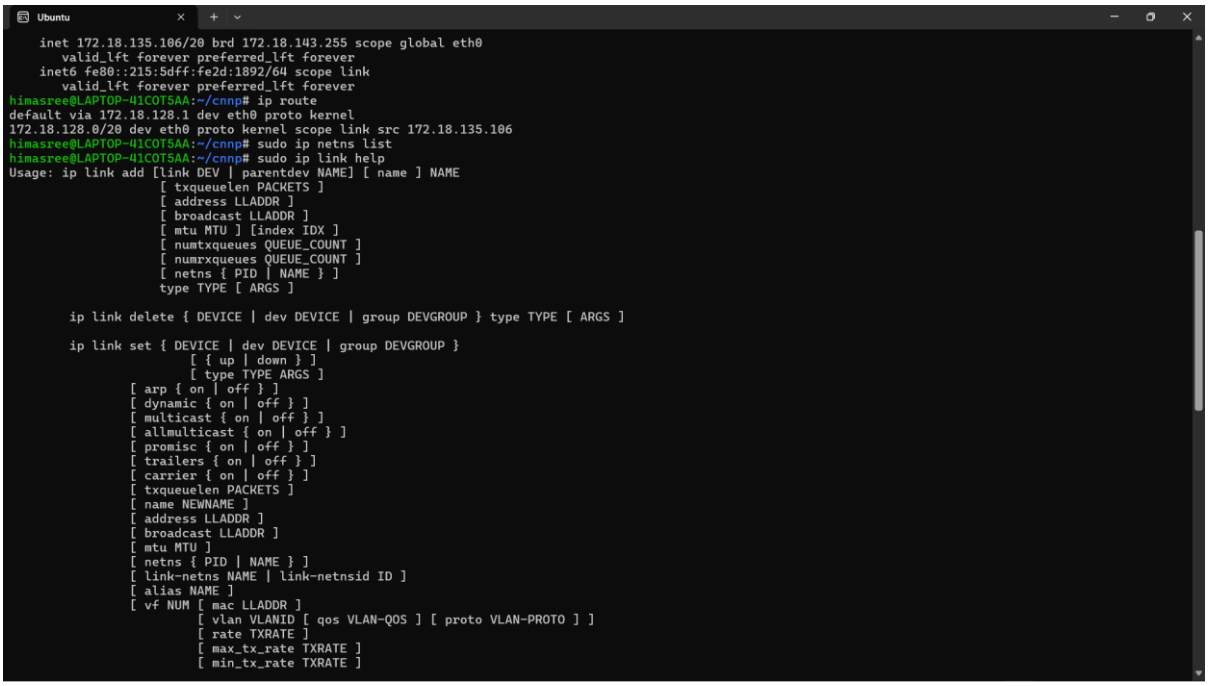
Skill – 3

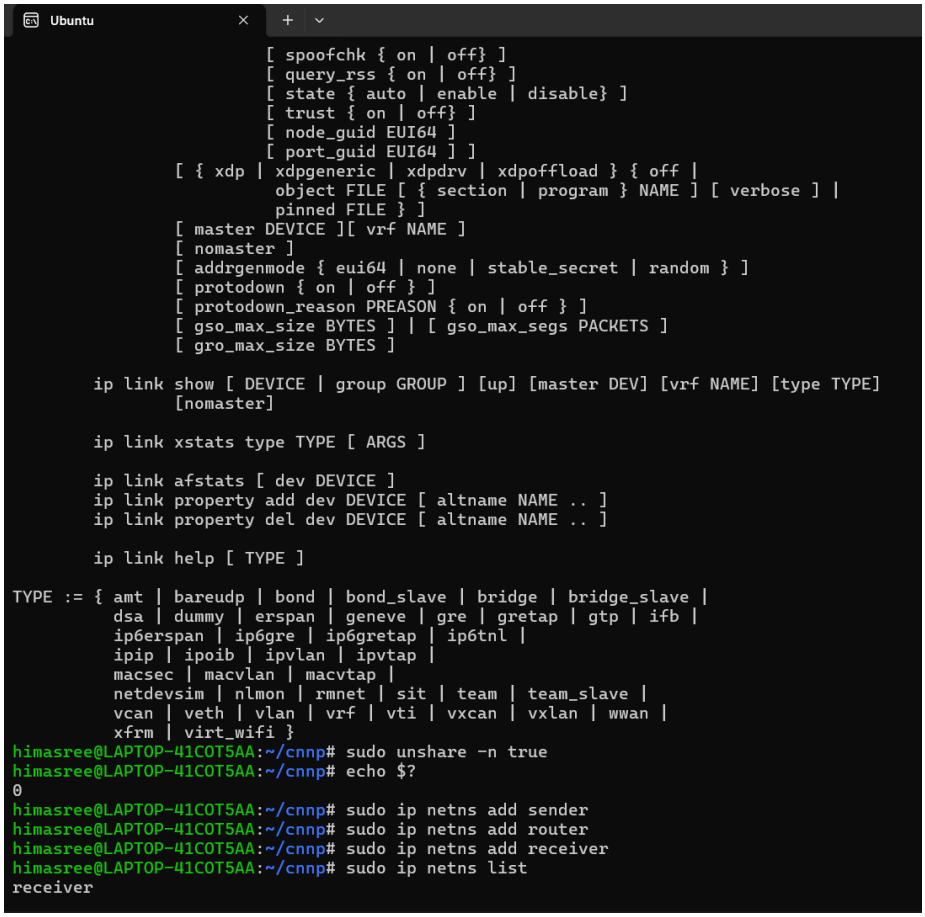
Name: Surapuraju Himasree
Roll No: 2520090181
Sec: 11

Step 1 – Install iPerf3 and Linux Traffic Control utilities





Step 2 – Configure the network




```
himasree@LAPTOP-41COT5AA:~/cnnp# sudo ip netns list
receiver
router
sender
himasree@LAPTOP-41COT5AA:~/cnnp# sudo ip link add sender0 type veth peer name router0
himasree@LAPTOP-41COT5AA:~/cnnp# sudo ip link add router1 type veth peer name receiver0
himasree@LAPTOP-41COT5AA:~/cnnp# sudo ip link set sender0 netns sender
himasree@LAPTOP-41COT5AA:~/cnnp# sudo ip link set router0 netns router
himasree@LAPTOP-41COT5AA:~/cnnp# sudo ip link set router1 netns router
himasree@LAPTOP-41COT5AA:~/cnnp# sudo ip link set receiver0 netns receiver
himasree@LAPTOP-41COT5AA:~/cnnp# sudo ip netns exec sender ip link
1: lo: <LOOPBACK> mtu 65536 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
4: sender0@if3: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/ether 22:f0:cf:f5:7f:72 brd ff:ff:ff:ff:ff:ff link-netns router
himasree@LAPTOP-41COT5AA:~/cnnp# sudo ip netns exec router ip link
1: lo: <LOOPBACK> mtu 65536 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
3: router0@if4: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/ether d2:cd:4d:2d:15:75 brd ff:ff:ff:ff:ff:ff link-netns sender
6: router1@if5: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/ether c2:4b:c5:79:bb:54 brd ff:ff:ff:ff:ff:ff link-netns receiver
himasree@LAPTOP-41COT5AA:~/cnnp# sudo ip netns exec receiver ip link
1: lo: <LOOPBACK> mtu 65536 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
5: receiver0@if6: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN mode DEFAULT group default qlen 1000
    link/ether 56:56:a6:2e:12:f0 brd ff:ff:ff:ff:ff:ff link-netns router
himasree@LAPTOP-41COT5AA:~/cnnp#
```

```
Ubuntu
       valid_lft forever preferred_lft forever
himasree@LAPTOP-41COT5AA:~/cnnp# sudo ip netns exec router sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip_forward = 1
himasree@LAPTOP-41COT5AA:~/cnnp# sudo ip netns exec sender ip route add 192.168.20.0/24 via 192.168.10.1
himasree@LAPTOP-41COT5AA:~/cnnp# sudo ip netns exec receiver ip route add 192.168.10.0/24 via 192.168.20.1
himasree@LAPTOP-41COT5AA:~/cnnp# sudo ip netns exec sender ip route
192.168.10.0/24 dev sender0 proto kernel scope link src 192.168.10.2
192.168.20.0/24 via 192.168.10.1 dev sender0
himasree@LAPTOP-41COT5AA:~/cnnp# sudo ip netns exec receiver ip route
192.168.10.0/24 via 192.168.20.1 dev receiver0
192.168.20.0/24 dev receiver0 proto kernel scope link src 192.168.20.2
himasree@LAPTOP-41COT5AA:~/cnnp# sudo ip netns exec sender ping -c 4 192.168.20.2
PING 192.168.20.2 (192.168.20.2) 56(84) bytes of data.
64 bytes from 192.168.20.2: icmp_seq=1 ttl=63 time=1.13 ms
64 bytes from 192.168.20.2: icmp_seq=2 ttl=63 time=0.073 ms
64 bytes from 192.168.20.2: icmp_seq=3 ttl=63 time=0.081 ms
64 bytes from 192.168.20.2: icmp_seq=4 ttl=63 time=0.068 ms

--- 192.168.20.2 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3056ms
rtt min/avg/max/mdev = 0.068/0.337/1.128/0.456 ms
```

Step 3 – Configure the bottleneck (5 Mbps)

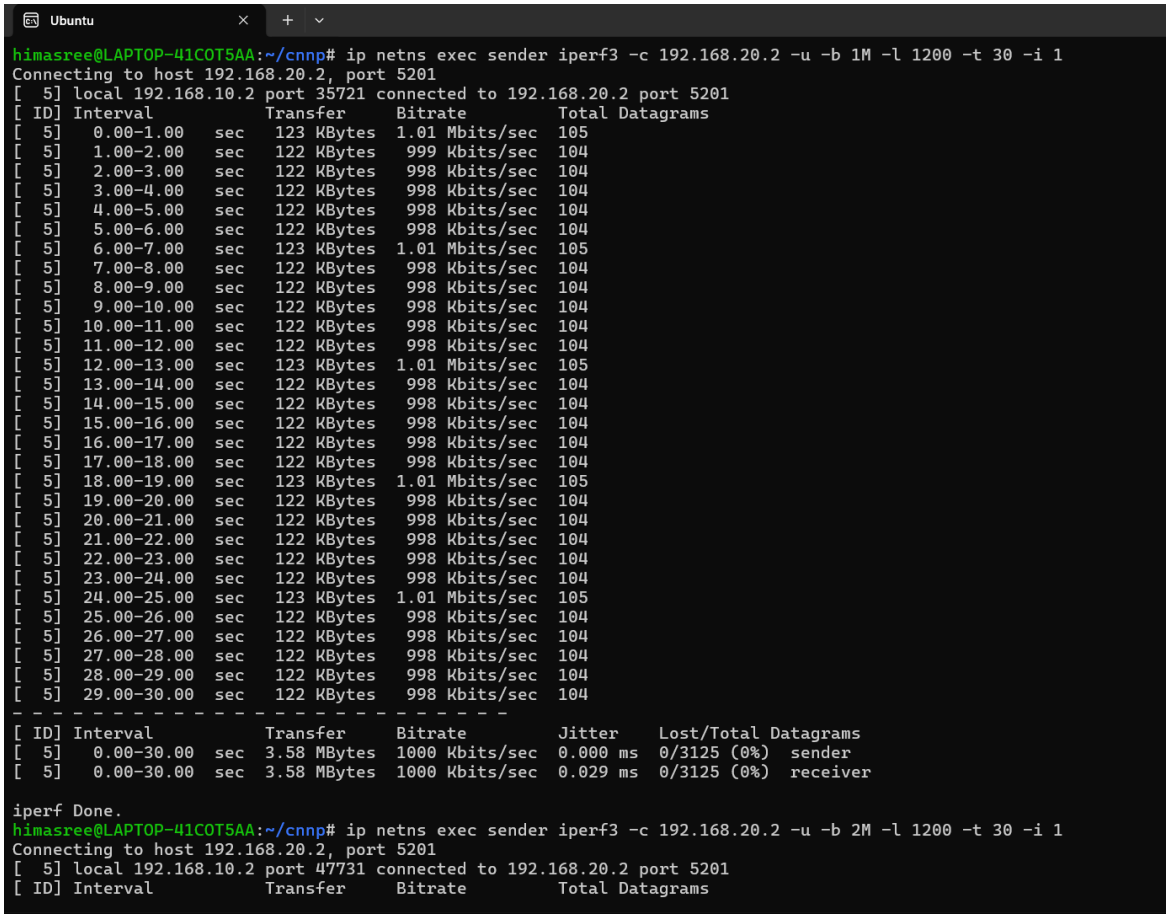
```
himasree@LAPTOP-41COT5AA:~/cnnp#   sudo ip netns exec router tc qdisc replace dev router1 root tbf rate 5mbit burst 32kb limit 64kb
himasree@LAPTOP-41COT5AA:~/cnnp# sudo ip netns exec router tc -s qdisc show dev router1
qdisc tbf 8001: root refcnt 17 rate 5Mbit burst 32Kb lat 52.4ms
 Sent 0 bytes 0 pkt (dropped 0, overlimits 0 requeues 0)
 backlog 0b 0p requeues 0
```

Step 4 – Start the iPerf3 receiver

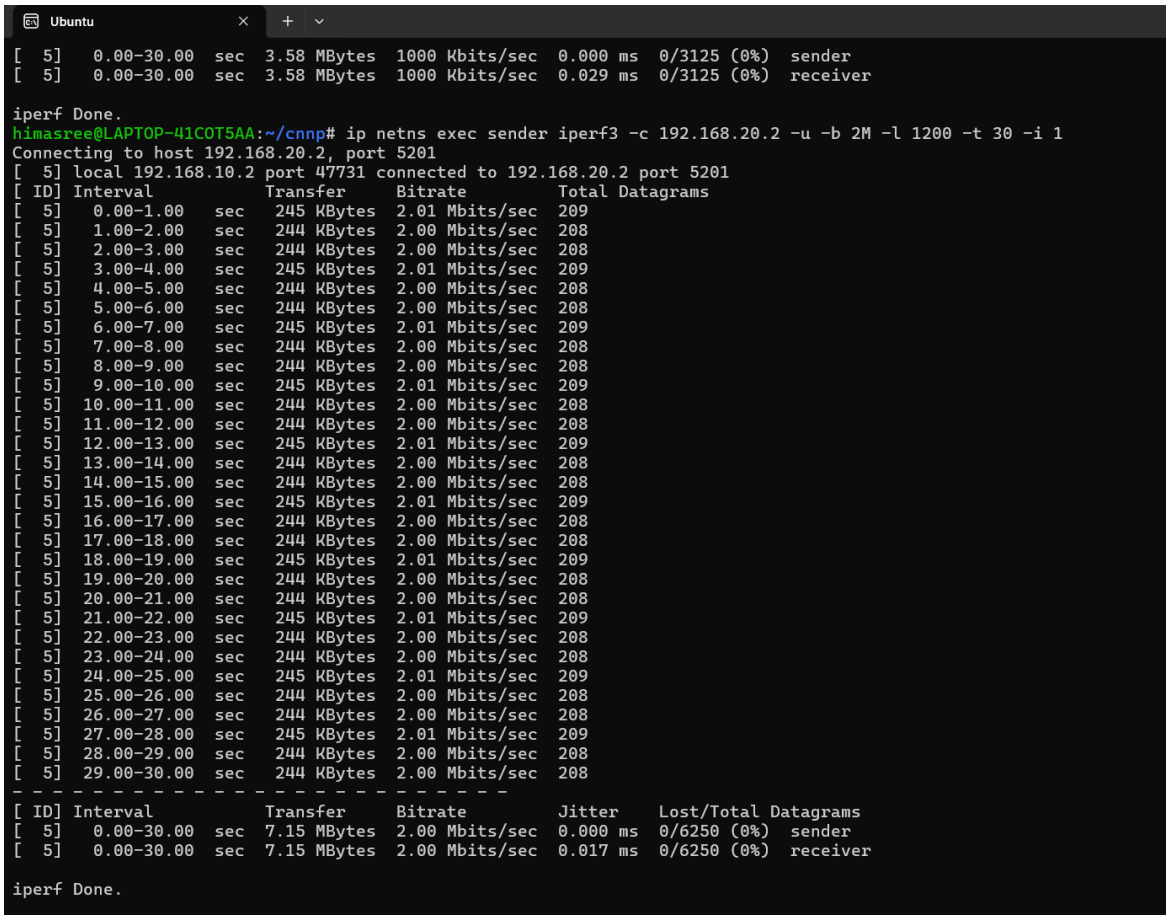
```
himasree@LAPTOP-41COT5AA:~/cnnp# sudo ip netns exec receiver iperf3 -s
-----------------------------------------------------------
Server listening on 5201 (test #1)
-----------------------------------------------------------
Accepted connection from 192.168.10.2, port 47178
[  5] local 192.168.20.2 port 5201 connected to 192.168.10.2 port 49681
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec   123 KBytes  1.01 Mbits/sec  0.098 ms  0/105 (0%)
[  5]   1.00-2.00   sec   122 KBytes   999 Kbits/sec  0.035 ms  0/104 (0%)
[  5]   2.00-3.00   sec   122 KBytes   998 Kbits/sec  0.064 ms  0/104 (0%)
[  5]   3.00-4.00   sec   122 KBytes   998 Kbits/sec  0.055 ms  0/104 (0%)
[  5]   4.00-5.00   sec   122 KBytes   998 Kbits/sec  0.041 ms  0/104 (0%)
[  5]   5.00-6.00   sec   122 KBytes   998 Kbits/sec  0.135 ms  0/104 (0%)
[  5]   6.00-7.00   sec   123 KBytes  1.01 Mbits/sec  0.051 ms  0/105 (0%)
[  5]   7.00-8.00   sec   122 KBytes   998 Kbits/sec  0.058 ms  0/104 (0%)
[  5]   8.00-9.00   sec   122 KBytes   998 Kbits/sec  0.058 ms  0/104 (0%)
[  5]   9.00-10.00  sec   122 KBytes   998 Kbits/sec  0.080 ms  0/104 (0%)
[  5]  10.00-11.00  sec   122 KBytes   998 Kbits/sec  0.044 ms  0/104 (0%)
```

Step 5 – Generate traffic from the sender

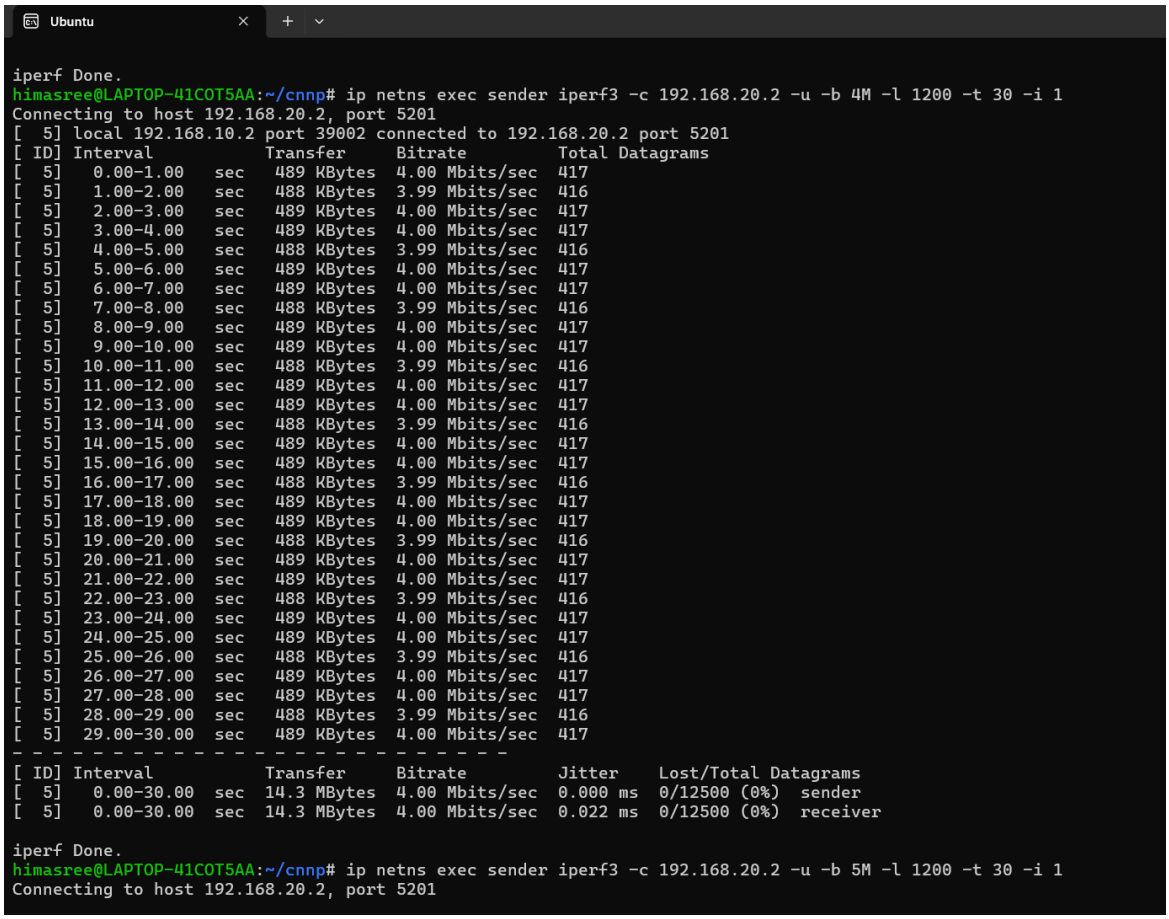
1 Mbps



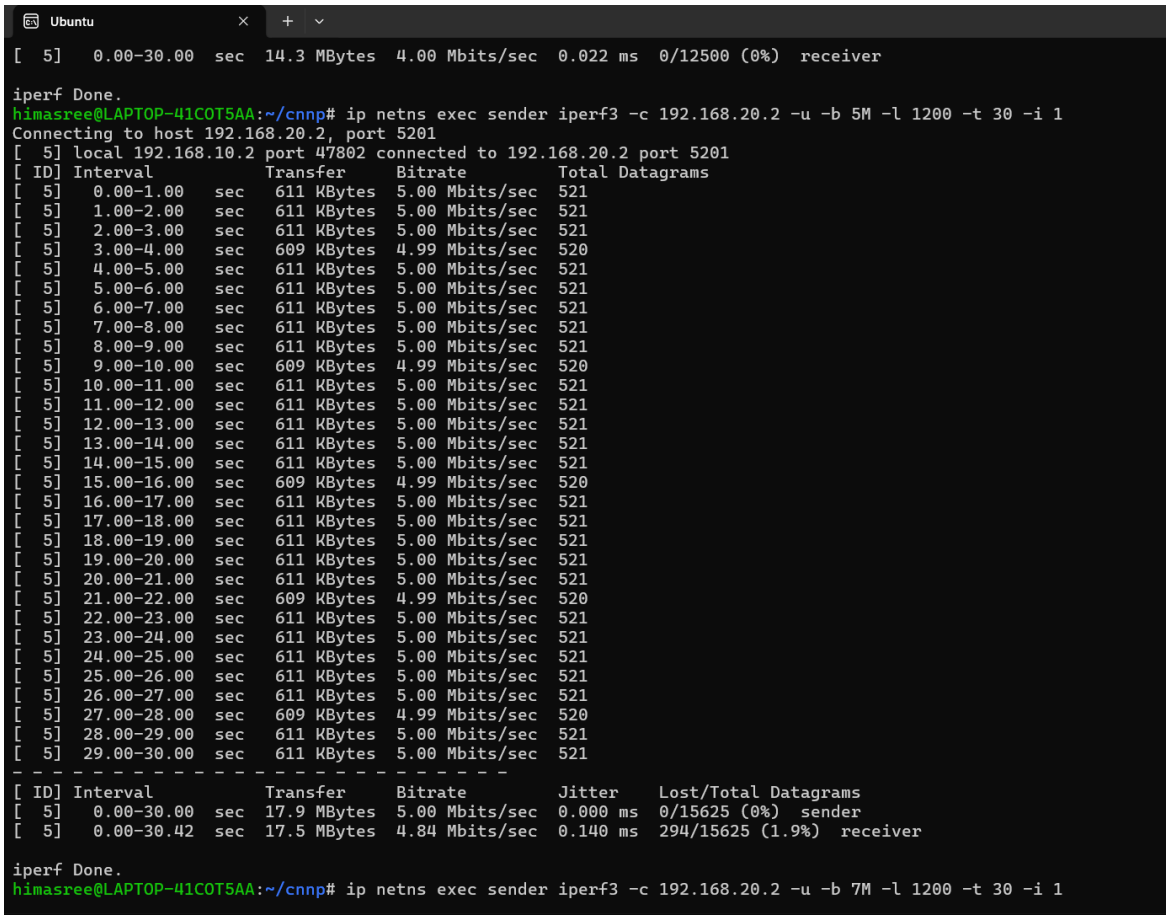
2 Mbps


Ubuntu
[5] 0.00-30.00 sec 3.58 MBytes 1000 Kbits/sec 0.000 ms 0/3125 (0%) sender
[5] 0.00-30.00 sec 3.58 MBytes 1000 Kbits/sec 0.029 ms 0/3125 (0%) receiver
iperf Done.
himasree@LAPTOP-41COT5AA:~/cnnp# ip netns exec sender iperf3 -c 192.168.20.2 -u -b 2M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
[5] local 192.168.10.2 port 47731 connected to 192.168.20.2 port 5201
[ID] Interval Transfer Bitrate Total Datagrams
[5] 0.00-1.00 sec 245 KBytes 2.01 Mbits/sec 209
[5] 1.00-2.00 sec 244 KBytes 2.00 Mbits/sec 208
[5] 2.00-3.00 sec 244 KBytes 2.00 Mbits/sec 208
[5] 3.00-4.00 sec 245 KBytes 2.01 Mbits/sec 209
[5] 4.00-5.00 sec 244 KBytes 2.00 Mbits/sec 208
[5] 5.00-6.00 sec 244 KBytes 2.00 Mbits/sec 208
[5] 6.00-7.00 sec 245 KBytes 2.01 Mbits/sec 209
[5] 7.00-8.00 sec 244 KBytes 2.00 Mbits/sec 208
[5] 8.00-9.00 sec 244 KBytes 2.00 Mbits/sec 208
[5] 9.00-10.00 sec 245 KBytes 2.01 Mbits/sec 209
[5] 10.00-11.00 sec 244 KBytes 2.00 Mbits/sec 208
[5] 11.00-12.00 sec 244 KBytes 2.00 Mbits/sec 208
[5] 12.00-13.00 sec 245 KBytes 2.01 Mbits/sec 209
[5] 13.00-14.00 sec 244 KBytes 2.00 Mbits/sec 208
[5] 14.00-15.00 sec 244 KBytes 2.00 Mbits/sec 208
[5] 15.00-16.00 sec 245 KBytes 2.01 Mbits/sec 209
[5] 16.00-17.00 sec 244 KBytes 2.00 Mbits/sec 208
[5] 17.00-18.00 sec 244 KBytes 2.00 Mbits/sec 208
[5] 18.00-19.00 sec 245 KBytes 2.01 Mbits/sec 209
[5] 19.00-20.00 sec 244 KBytes 2.00 Mbits/sec 208
[5] 20.00-21.00 sec 244 KBytes 2.00 Mbits/sec 208
[5] 21.00-22.00 sec 245 KBytes 2.01 Mbits/sec 209
[5] 22.00-23.00 sec 244 KBytes 2.00 Mbits/sec 208
[5] 23.00-24.00 sec 244 KBytes 2.00 Mbits/sec 208
[5] 24.00-25.00 sec 245 KBytes 2.01 Mbits/sec 209
[5] 25.00-26.00 sec 244 KBytes 2.00 Mbits/sec 208
[5] 26.00-27.00 sec 244 KBytes 2.00 Mbits/sec 208
[5] 27.00-28.00 sec 245 KBytes 2.01 Mbits/sec 209
[5] 28.00-29.00 sec 244 KBytes 2.00 Mbits/sec 208
[5] 29.00-30.00 sec 244 KBytes 2.00 Mbits/sec 208
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-30.00 sec 7.15 MBytes 2.00 Mbits/sec 0.000 ms 0/6250 (0%) sender
[5] 0.00-30.00 sec 7.15 MBytes 2.00 Mbits/sec 0.017 ms 0/6250 (0%) receiver
iperf Done.

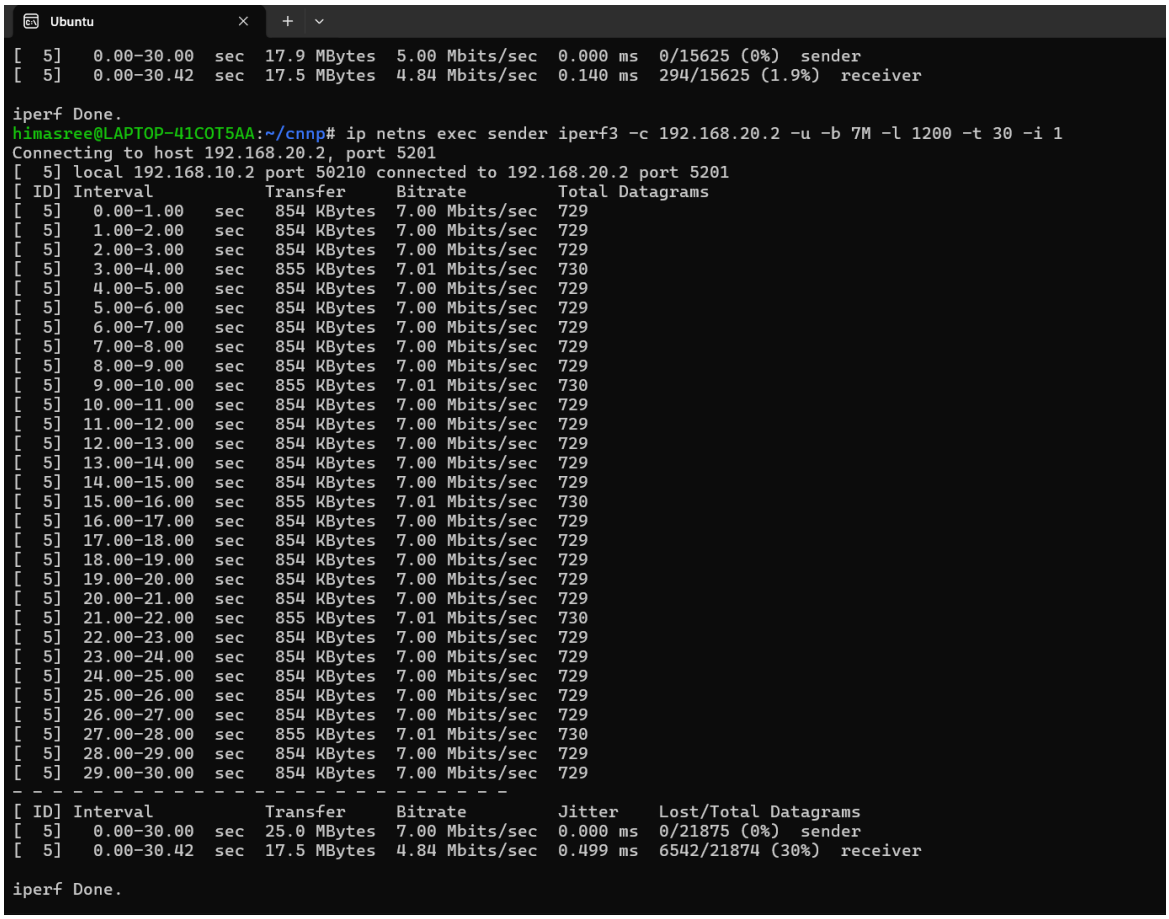
4 Mbps


Ubuntu
iperf Done.
himasree@LAPTOP-41COT5AA:~/cnnp# ip netns exec sender iperf3 -c 192.168.20.2 -u -b 4M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
[5] local 192.168.10.2 port 39002 connected to 192.168.20.2 port 5201
[ID] Interval Transfer Bitrate Total Datagrams
[5] 0.00-1.00 sec 489 KBytes 4.00 Mbits/sec 417
[5] 1.00-2.00 sec 488 KBytes 3.99 Mbits/sec 416
[5] 2.00-3.00 sec 489 KBytes 4.00 Mbits/sec 417
[5] 3.00-4.00 sec 489 KBytes 4.00 Mbits/sec 417
[5] 4.00-5.00 sec 488 KBytes 3.99 Mbits/sec 416
[5] 5.00-6.00 sec 489 KBytes 4.00 Mbits/sec 417
[5] 6.00-7.00 sec 489 KBytes 4.00 Mbits/sec 417
[5] 7.00-8.00 sec 488 KBytes 3.99 Mbits/sec 416
[5] 8.00-9.00 sec 489 KBytes 4.00 Mbits/sec 417
[5] 9.00-10.00 sec 489 KBytes 4.00 Mbits/sec 417
[5] 10.00-11.00 sec 488 KBytes 3.99 Mbits/sec 416
[5] 11.00-12.00 sec 489 KBytes 4.00 Mbits/sec 417
[5] 12.00-13.00 sec 489 KBytes 4.00 Mbits/sec 417
[5] 13.00-14.00 sec 488 KBytes 3.99 Mbits/sec 416
[5] 14.00-15.00 sec 489 KBytes 4.00 Mbits/sec 417
[5] 15.00-16.00 sec 489 KBytes 4.00 Mbits/sec 417
[5] 16.00-17.00 sec 488 KBytes 3.99 Mbits/sec 416
[5] 17.00-18.00 sec 489 KBytes 4.00 Mbits/sec 417
[5] 18.00-19.00 sec 489 KBytes 4.00 Mbits/sec 417
[5] 19.00-20.00 sec 488 KBytes 3.99 Mbits/sec 416
[5] 20.00-21.00 sec 489 KBytes 4.00 Mbits/sec 417
[5] 21.00-22.00 sec 489 KBytes 4.00 Mbits/sec 417
[5] 22.00-23.00 sec 488 KBytes 3.99 Mbits/sec 416
[5] 23.00-24.00 sec 489 KBytes 4.00 Mbits/sec 417
[5] 24.00-25.00 sec 489 KBytes 4.00 Mbits/sec 417
[5] 25.00-26.00 sec 488 KBytes 3.99 Mbits/sec 416
[5] 26.00-27.00 sec 489 KBytes 4.00 Mbits/sec 417
[5] 27.00-28.00 sec 489 KBytes 4.00 Mbits/sec 417
[5] 28.00-29.00 sec 488 KBytes 3.99 Mbits/sec 416
[5] 29.00-30.00 sec 489 KBytes 4.00 Mbits/sec 417
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-30.00 sec 14.3 MBytes 4.00 Mbits/sec 0.000 ms 0/12500 (0%) sender
[5] 0.00-30.00 sec 14.3 MBytes 4.00 Mbits/sec 0.022 ms 0/12500 (0%) receiver
iperf Done.
himasree@LAPTOP-41COT5AA:~/cnnp# ip netns exec sender iperf3 -c 192.168.20.2 -u -b 5M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201

5 Mbps


Ubuntu
[5] 0.00-30.00 sec 14.3 MBytes 4.00 Mbits/sec 0.022 ms 0/12500 (0%) receiver
iperf Done.
himasree@LAPTOP-41COT5AA:~/cnnp# ip netns exec sender iperf3 -c 192.168.20.2 -u -b 5M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
[5] local 192.168.10.2 port 47802 connected to 192.168.20.2 port 5201
[ID] Interval Transfer Bitrate Total Datagrams
[5] 0.00-1.00 sec 611 KBytes 5.00 Mbits/sec 521
[5] 1.00-2.00 sec 611 KBytes 5.00 Mbits/sec 521
[5] 2.00-3.00 sec 611 KBytes 5.00 Mbits/sec 521
[5] 3.00-4.00 sec 609 KBytes 4.99 Mbits/sec 520
[5] 4.00-5.00 sec 611 KBytes 5.00 Mbits/sec 521
[5] 5.00-6.00 sec 611 KBytes 5.00 Mbits/sec 521
[5] 6.00-7.00 sec 611 KBytes 5.00 Mbits/sec 521
[5] 7.00-8.00 sec 611 KBytes 5.00 Mbits/sec 521
[5] 8.00-9.00 sec 611 KBytes 5.00 Mbits/sec 521
[5] 9.00-10.00 sec 609 KBytes 4.99 Mbits/sec 520
[5] 10.00-11.00 sec 611 KBytes 5.00 Mbits/sec 521
[5] 11.00-12.00 sec 611 KBytes 5.00 Mbits/sec 521
[5] 12.00-13.00 sec 611 KBytes 5.00 Mbits/sec 521
[5] 13.00-14.00 sec 611 KBytes 5.00 Mbits/sec 521
[5] 14.00-15.00 sec 611 KBytes 5.00 Mbits/sec 521
[5] 15.00-16.00 sec 609 KBytes 4.99 Mbits/sec 520
[5] 16.00-17.00 sec 611 KBytes 5.00 Mbits/sec 521
[5] 17.00-18.00 sec 611 KBytes 5.00 Mbits/sec 521
[5] 18.00-19.00 sec 611 KBytes 5.00 Mbits/sec 521
[5] 19.00-20.00 sec 611 KBytes 5.00 Mbits/sec 521
[5] 20.00-21.00 sec 611 KBytes 5.00 Mbits/sec 521
[5] 21.00-22.00 sec 609 KBytes 4.99 Mbits/sec 520
[5] 22.00-23.00 sec 611 KBytes 5.00 Mbits/sec 521
[5] 23.00-24.00 sec 611 KBytes 5.00 Mbits/sec 521
[5] 24.00-25.00 sec 611 KBytes 5.00 Mbits/sec 521
[5] 25.00-26.00 sec 611 KBytes 5.00 Mbits/sec 521
[5] 26.00-27.00 sec 611 KBytes 5.00 Mbits/sec 521
[5] 27.00-28.00 sec 609 KBytes 4.99 Mbits/sec 520
[5] 28.00-29.00 sec 611 KBytes 5.00 Mbits/sec 521
[5] 29.00-30.00 sec 611 KBytes 5.00 Mbits/sec 521
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-30.00 sec 17.9 MBytes 5.00 Mbits/sec 0.000 ms 0/15625 (0%) sender
[5] 0.00-30.42 sec 17.5 MBytes 4.84 Mbits/sec 0.140 ms 294/15625 (1.9%) receiver
iperf Done.
himasree@LAPTOP-41COT5AA:~/cnnp# ip netns exec sender iperf3 -c 192.168.20.2 -u -b 7M -l 1200 -t 30 -i 1

7 Mbps


Ubuntu
[5] 0.00-30.00 sec 17.9 MBytes 5.00 Mbits/sec 0.000 ms 0/15625 (0%) sender
[5] 0.00-30.42 sec 17.5 MBytes 4.84 Mbits/sec 0.140 ms 294/15625 (1.9%) receiver
iperf Done.
himasree@LAPTOP-41COT5AA:~/cnnp# ip netns exec sender iperf3 -c 192.168.20.2 -u -b 7M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
[5] local 192.168.10.2 port 50210 connected to 192.168.20.2 port 5201
[ID] Interval Transfer Bitrate Total Datagrams
[5] 0.00-1.00 sec 854 KBytes 7.00 Mbits/sec 729
[5] 1.00-2.00 sec 854 KBytes 7.00 Mbits/sec 729
[5] 2.00-3.00 sec 854 KBytes 7.00 Mbits/sec 729
[5] 3.00-4.00 sec 855 KBytes 7.01 Mbits/sec 730
[5] 4.00-5.00 sec 854 KBytes 7.00 Mbits/sec 729
[5] 5.00-6.00 sec 854 KBytes 7.00 Mbits/sec 729
[5] 6.00-7.00 sec 854 KBytes 7.00 Mbits/sec 729
[5] 7.00-8.00 sec 854 KBytes 7.00 Mbits/sec 729
[5] 8.00-9.00 sec 854 KBytes 7.00 Mbits/sec 729
[5] 9.00-10.00 sec 855 KBytes 7.01 Mbits/sec 730
[5] 10.00-11.00 sec 854 KBytes 7.00 Mbits/sec 729
[5] 11.00-12.00 sec 854 KBytes 7.00 Mbits/sec 729
[5] 12.00-13.00 sec 854 KBytes 7.00 Mbits/sec 729
[5] 13.00-14.00 sec 854 KBytes 7.00 Mbits/sec 729
[5] 14.00-15.00 sec 854 KBytes 7.00 Mbits/sec 729
[5] 15.00-16.00 sec 855 KBytes 7.01 Mbits/sec 730
[5] 16.00-17.00 sec 854 KBytes 7.00 Mbits/sec 729
[5] 17.00-18.00 sec 854 KBytes 7.00 Mbits/sec 729
[5] 18.00-19.00 sec 854 KBytes 7.00 Mbits/sec 729
[5] 19.00-20.00 sec 854 KBytes 7.00 Mbits/sec 729
[5] 20.00-21.00 sec 854 KBytes 7.00 Mbits/sec 729
[5] 21.00-22.00 sec 855 KBytes 7.01 Mbits/sec 730
[5] 22.00-23.00 sec 854 KBytes 7.00 Mbits/sec 729
[5] 23.00-24.00 sec 854 KBytes 7.00 Mbits/sec 729
[5] 24.00-25.00 sec 854 KBytes 7.00 Mbits/sec 729
[5] 25.00-26.00 sec 854 KBytes 7.00 Mbits/sec 729
[5] 26.00-27.00 sec 854 KBytes 7.00 Mbits/sec 729
[5] 27.00-28.00 sec 855 KBytes 7.01 Mbits/sec 730
[5] 28.00-29.00 sec 854 KBytes 7.00 Mbits/sec 729
[5] 29.00-30.00 sec 854 KBytes 7.00 Mbits/sec 729
[ID] Interval Transfer Bitrate Jitter Lost/Total Datagrams
[5] 0.00-30.00 sec 25.0 MBytes 7.00 Mbits/sec 0.000 ms 0/21875 (0%) sender
[5] 0.00-30.42 sec 17.5 MBytes 4.84 Mbits/sec 0.499 ms 6542/21874 (30%) receiver
iperf Done.

10 Mbps

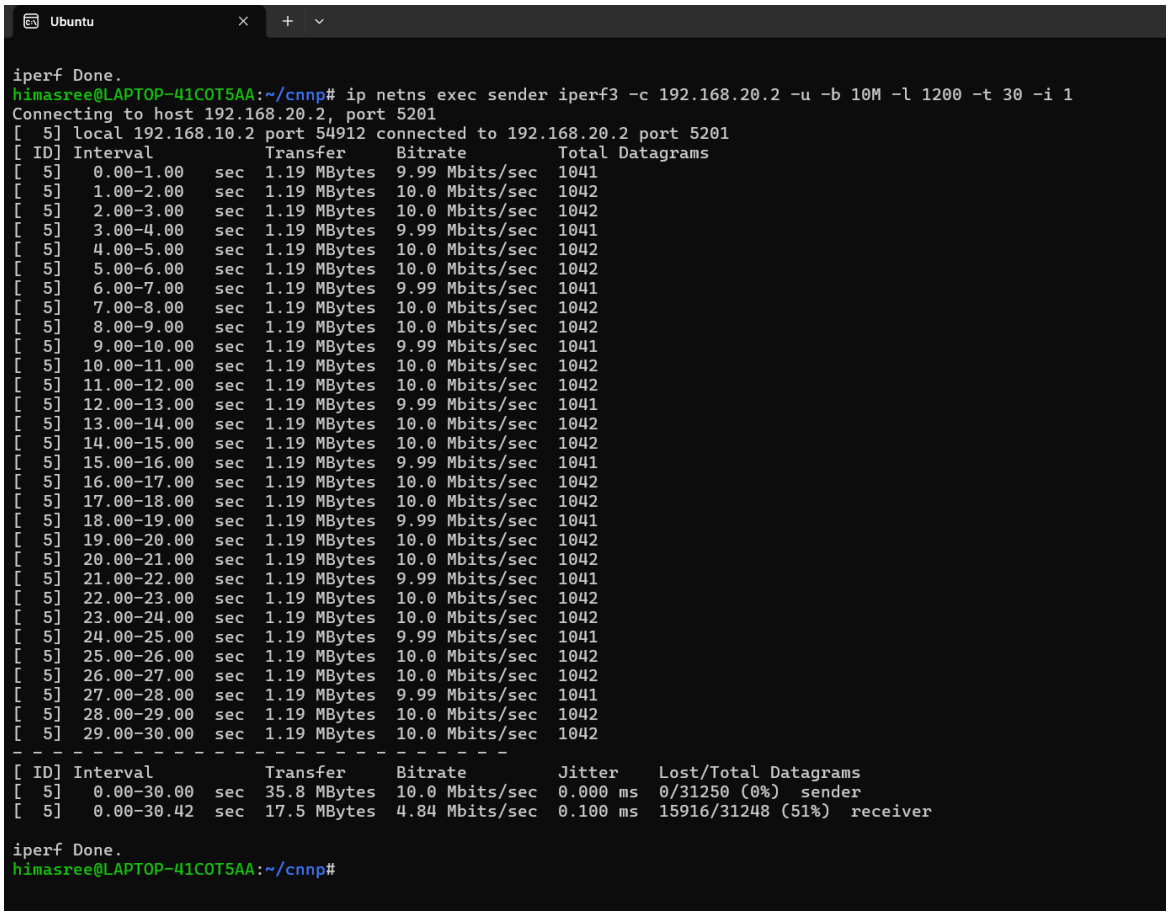


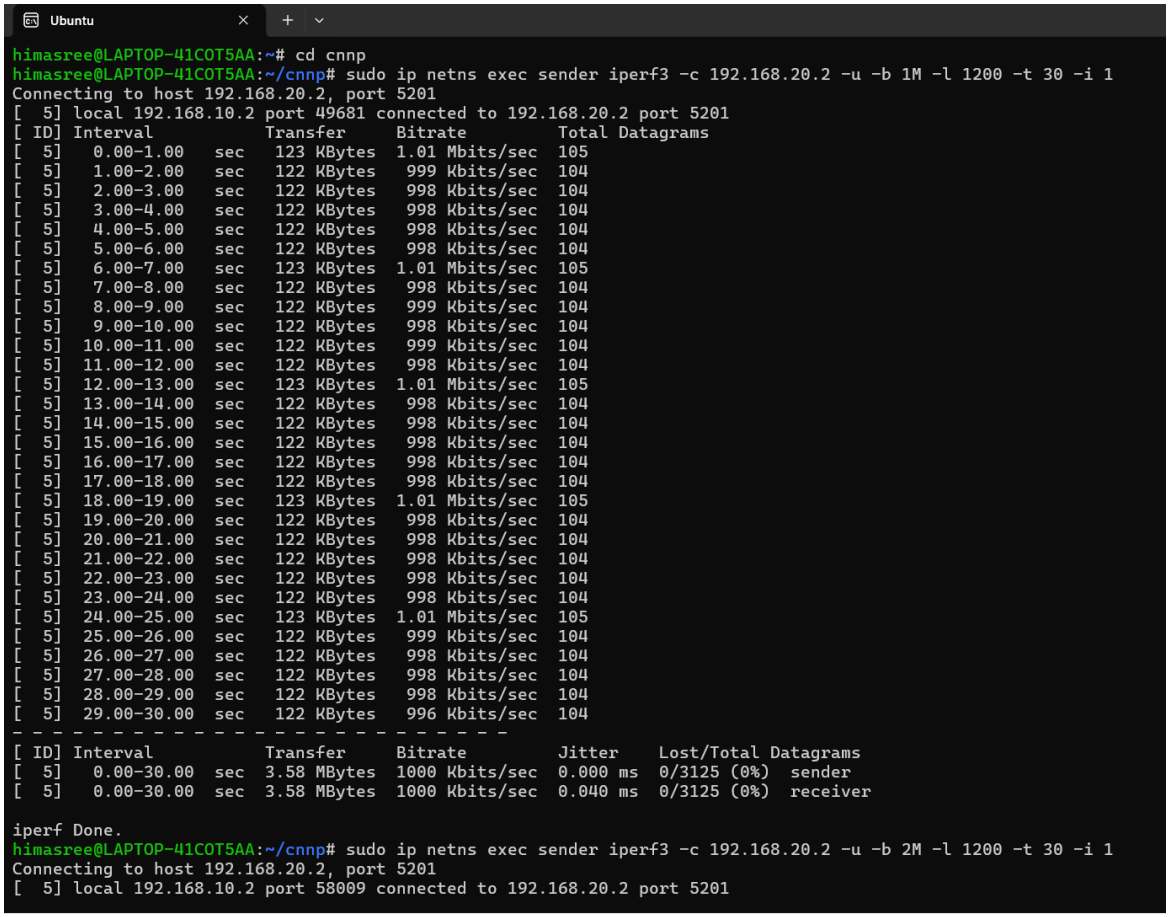
Table 1: Traffic rate results

Offered traffic (Mbps)	Receiver throughput (Mbps)	Packet loss (%)	Total datagrams	Lost datagrams	Received datagrams	Jitter (ms)
1	1.00	0	3125	0	3125	0.059
2	2.00	0	6250	0	6250	0.022
4	4.00	0	12500	0	12500	0.022
5	4.84	1.9	15625	294	15331	0.140
7	4.84	30	21874	6542	15332	0.499
10	4.84	51	31248	15916	15332	0.100

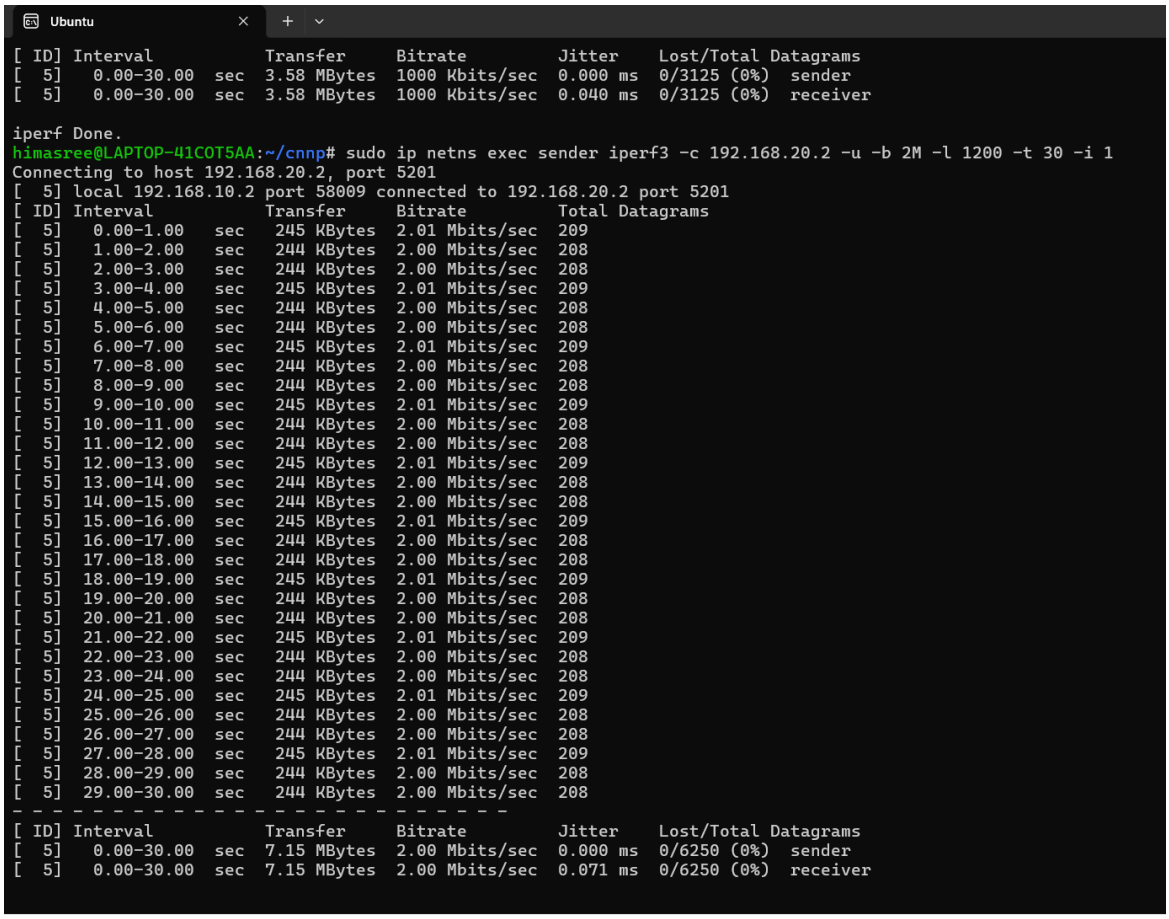
Packet loss = lost / total × 100. At 5 Mbps: 294 / 15625 × 100 = 1.9%. At 7 Mbps: 6542 / 21874 × 100 = 30%. At 10 Mbps: 15916 / 31248 × 100 = 51%.

Earlier run (not used in the table above):

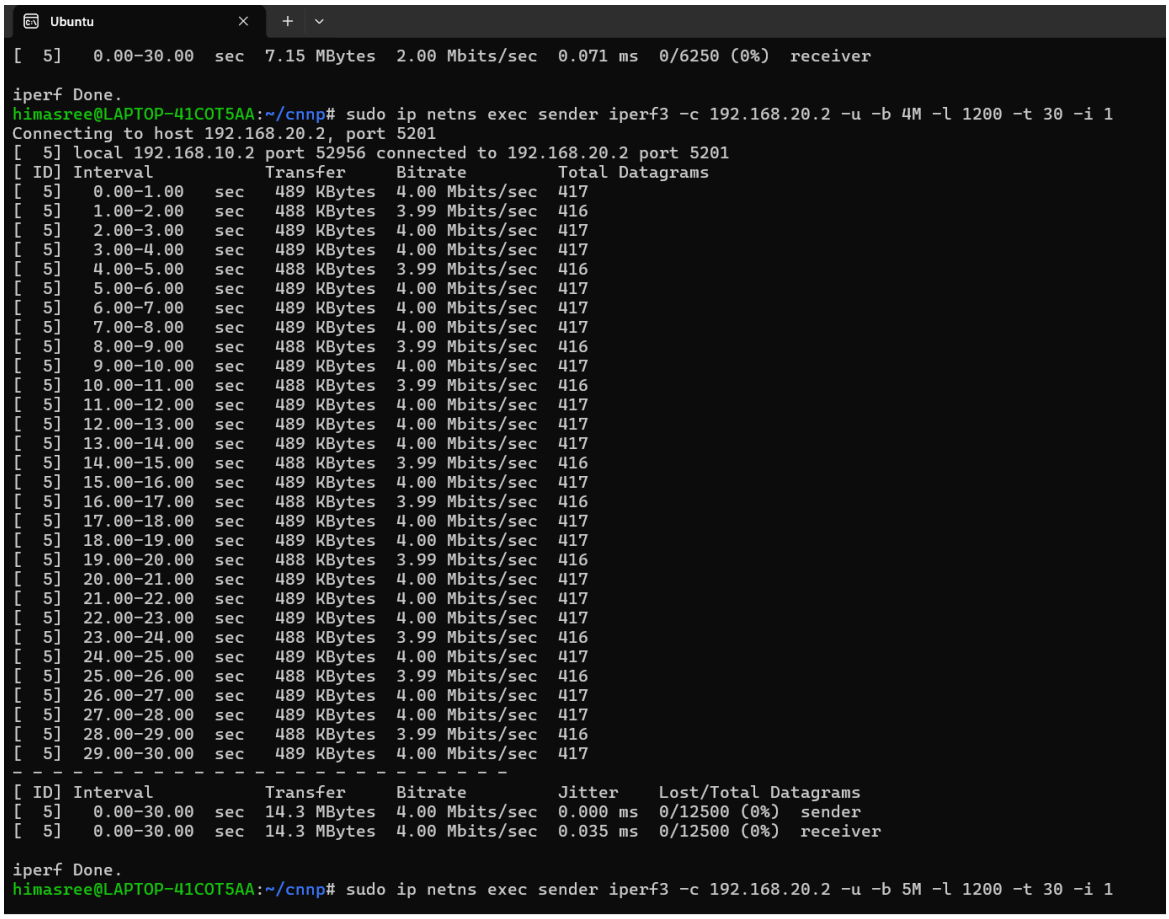
1 Mbps (earlier run)



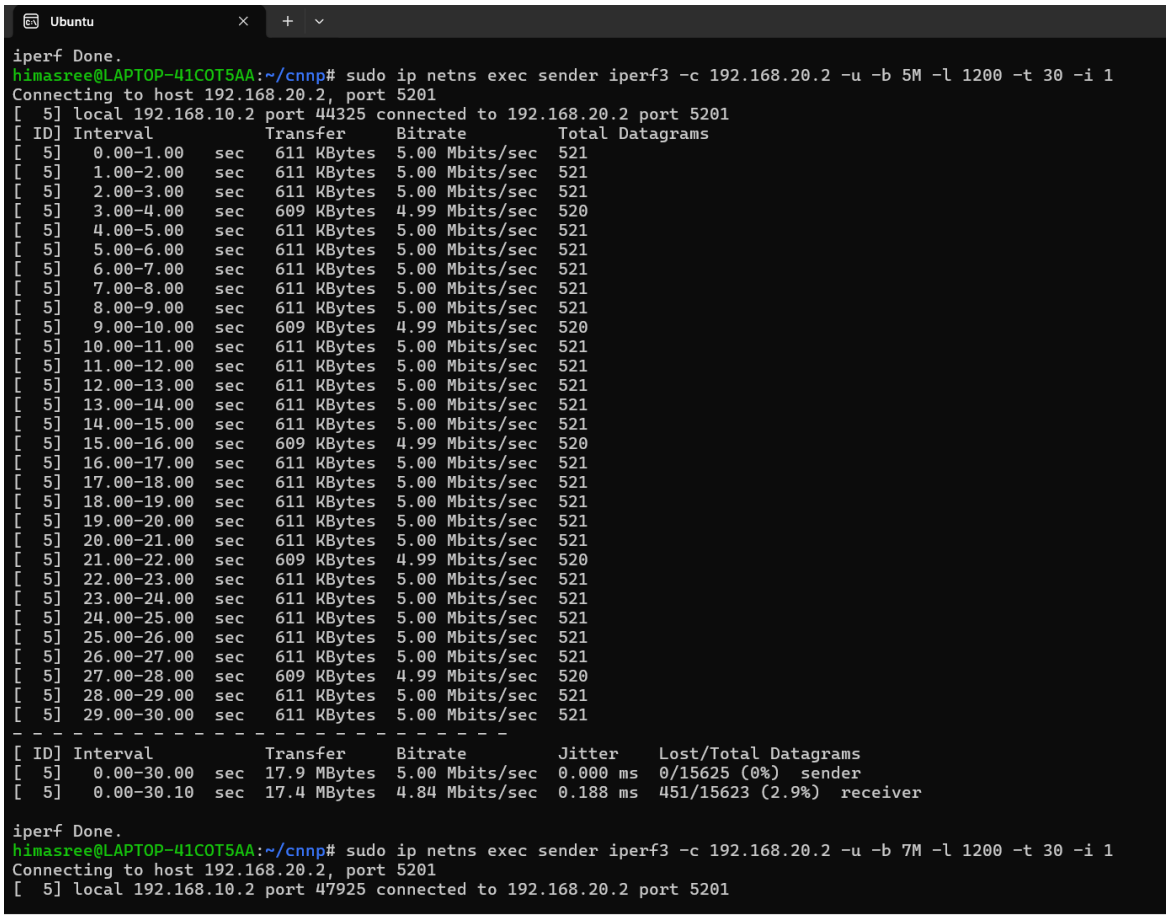
2 Mbps (earlier run)



4 Mbps (earlier run)



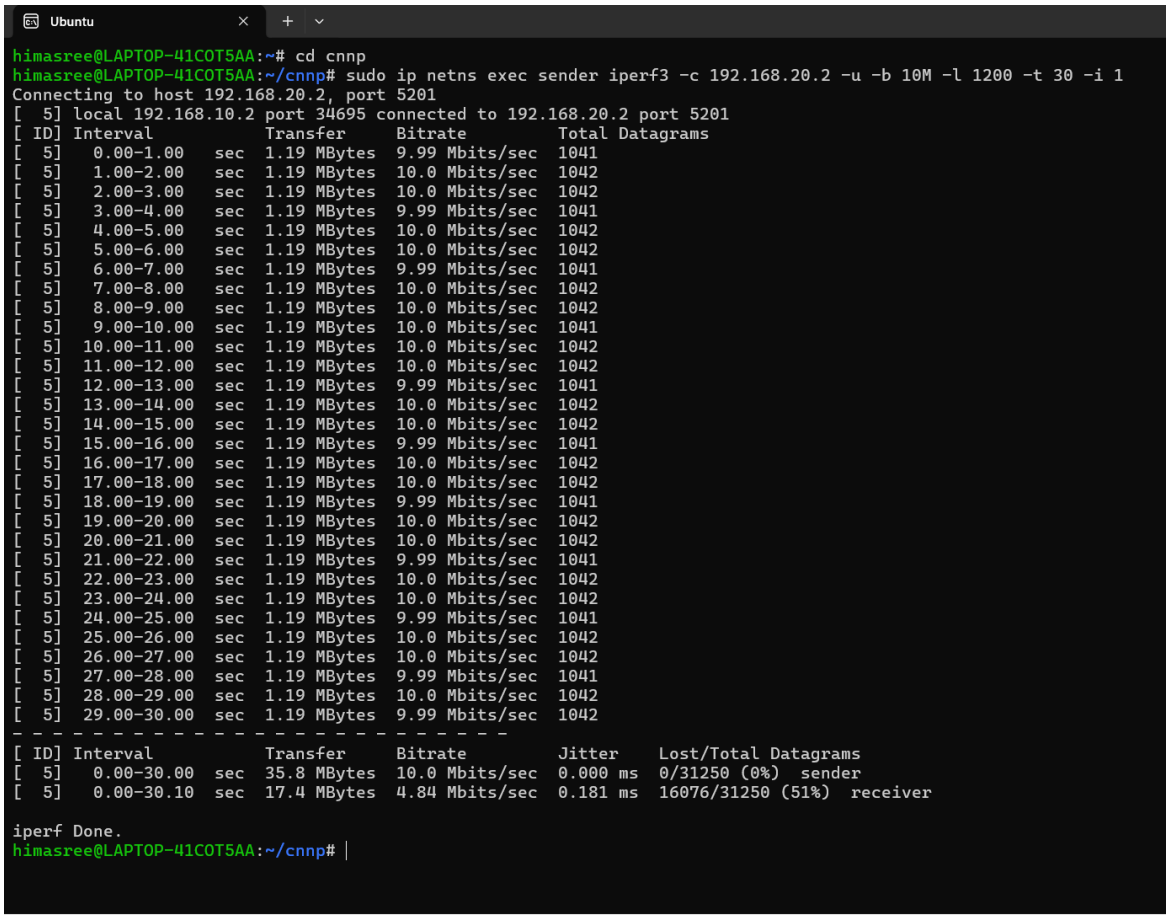
5 Mbps (earlier run)



```
Ubuntu
iperf Done.
himasree@LAPTOP-41COT5AA:~/cnnp# sudo ip netns exec sender iperf3 -c 192.168.20.2 -u -b 5M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 44325 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   611 KBytes  5.00 Mbits/sec  521
[  5]   1.00-2.00   sec   611 KBytes  5.00 Mbits/sec  521
[  5]   2.00-3.00   sec   611 KBytes  5.00 Mbits/sec  521
[  5]   3.00-4.00   sec   609 KBytes  4.99 Mbits/sec  520
[  5]   4.00-5.00   sec   611 KBytes  5.00 Mbits/sec  521
[  5]   5.00-6.00   sec   611 KBytes  5.00 Mbits/sec  521
[  5]   6.00-7.00   sec   611 KBytes  5.00 Mbits/sec  521
[  5]   7.00-8.00   sec   611 KBytes  5.00 Mbits/sec  521
[  5]   8.00-9.00   sec   611 KBytes  5.00 Mbits/sec  521
[  5]   9.00-10.00  sec   609 KBytes  4.99 Mbits/sec  520
[  5]  10.00-11.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  11.00-12.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  12.00-13.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  13.00-14.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  14.00-15.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  15.00-16.00  sec   609 KBytes  4.99 Mbits/sec  520
[  5]  16.00-17.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  17.00-18.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  18.00-19.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  19.00-20.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  20.00-21.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  21.00-22.00  sec   609 KBytes  4.99 Mbits/sec  520
[  5]  22.00-23.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  23.00-24.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  24.00-25.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  25.00-26.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  26.00-27.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  27.00-28.00  sec   609 KBytes  4.99 Mbits/sec  520
[  5]  28.00-29.00  sec   611 KBytes  5.00 Mbits/sec  521
[  5]  29.00-30.00  sec   611 KBytes  5.00 Mbits/sec  521
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-30.00  sec  17.9 MBytes  5.00 Mbits/sec  0.000 ms  0/15625 (0%)  sender
[  5]   0.00-30.10  sec  17.4 MBytes  4.84 Mbits/sec  0.188 ms  451/15623 (2.9%)  receiver

iperf Done.
himasree@LAPTOP-41COT5AA:~/cnnp# sudo ip netns exec sender iperf3 -c 192.168.20.2 -u -b 7M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 47925 connected to 192.168.20.2 port 5201
```

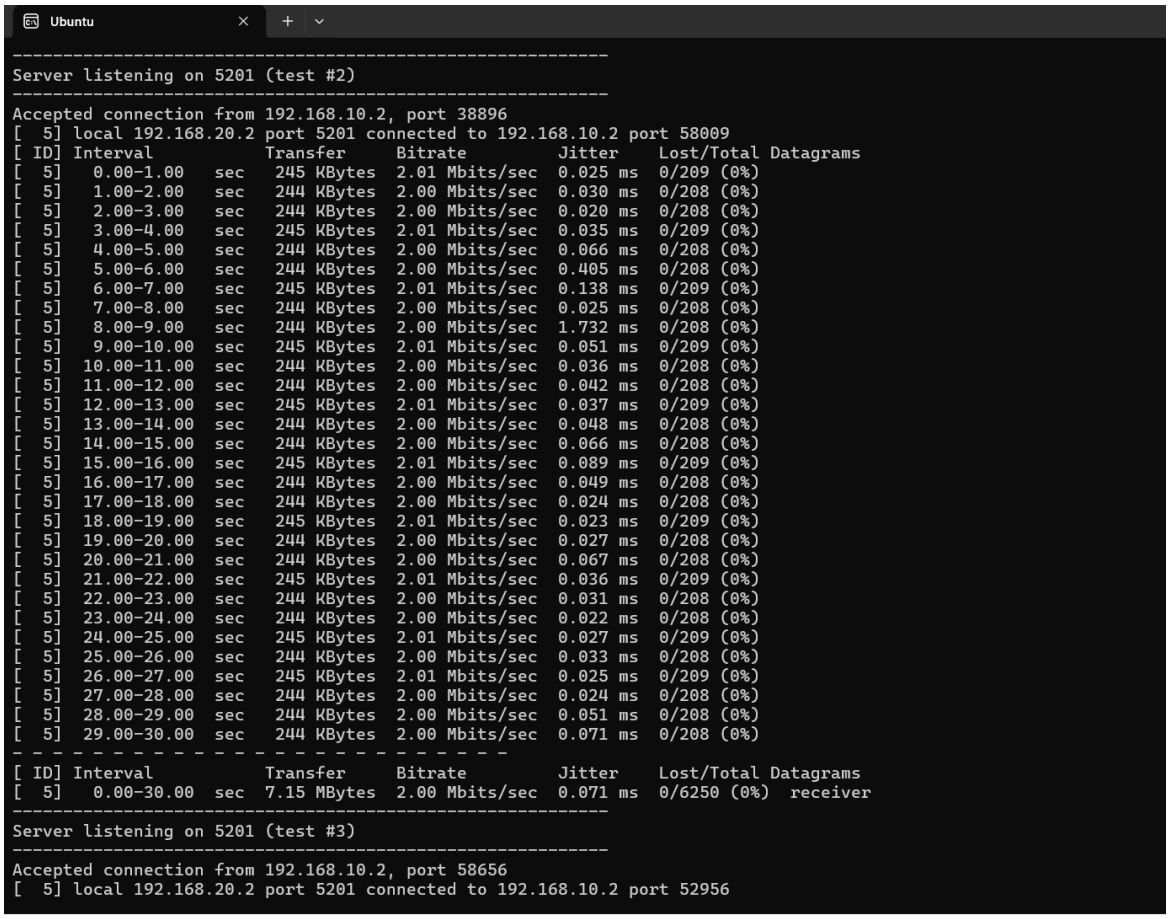
10 Mbps (earlier run)



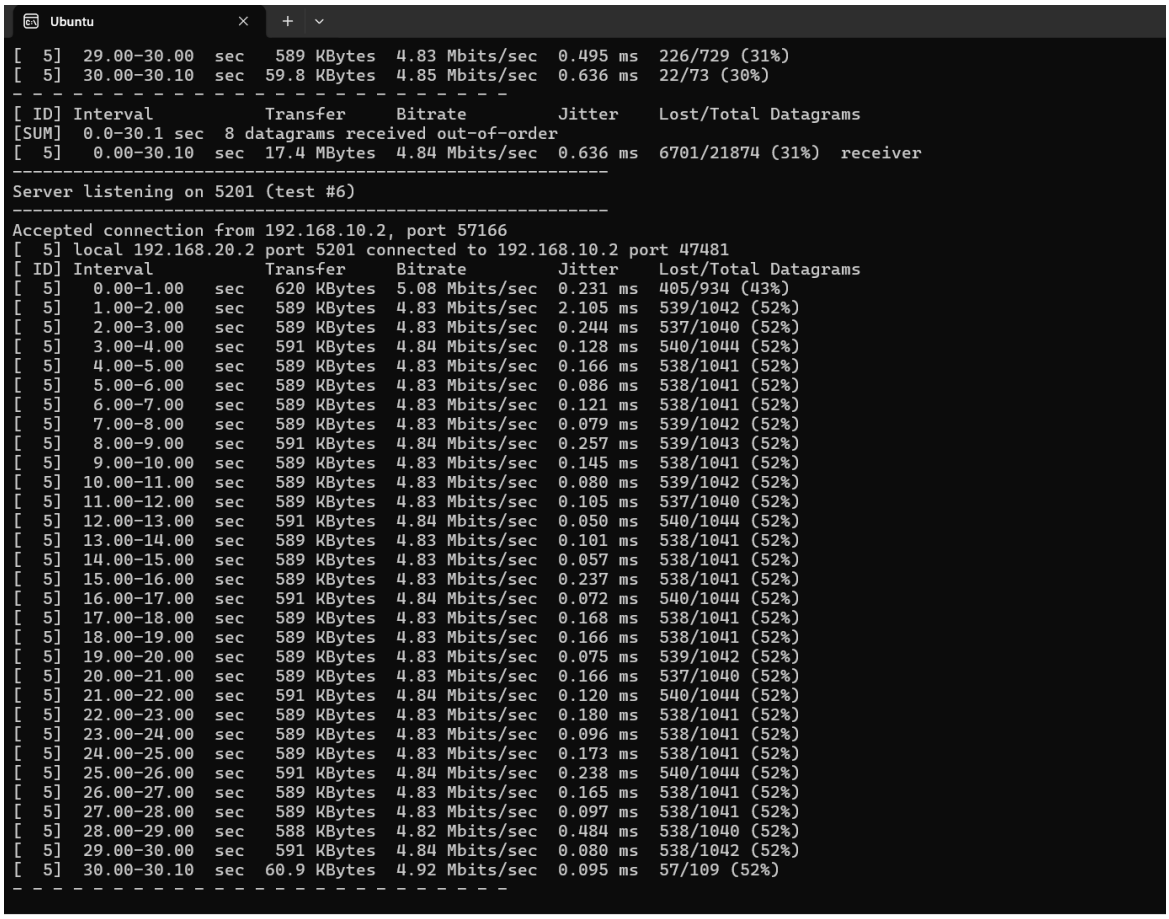
```
Ubuntu
himasree@LAPTOP-41COT5AA:~# cd cnnp
himasree@LAPTOP-41COT5AA:~/cnnp# sudo ip netns exec sender iperf3 -c 192.168.20.2 -u -b 10M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 34695 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec  1.19 MBytes  9.99 Mbits/sec  1041
[  5]   1.00-2.00   sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]   2.00-3.00   sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]   3.00-4.00   sec  1.19 MBytes  9.99 Mbits/sec  1041
[  5]   4.00-5.00   sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]   5.00-6.00   sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]   6.00-7.00   sec  1.19 MBytes  9.99 Mbits/sec  1041
[  5]   7.00-8.00   sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]   8.00-9.00   sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]   9.00-10.00  sec  1.19 MBytes  10.0 Mbits/sec  1041
[  5]  10.00-11.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  11.00-12.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  12.00-13.00  sec  1.19 MBytes  9.99 Mbits/sec  1041
[  5]  13.00-14.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  14.00-15.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  15.00-16.00  sec  1.19 MBytes  9.99 Mbits/sec  1041
[  5]  16.00-17.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  17.00-18.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  18.00-19.00  sec  1.19 MBytes  9.99 Mbits/sec  1041
[  5]  19.00-20.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  20.00-21.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  21.00-22.00  sec  1.19 MBytes  9.99 Mbits/sec  1041
[  5]  22.00-23.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  23.00-24.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  24.00-25.00  sec  1.19 MBytes  9.99 Mbits/sec  1041
[  5]  25.00-26.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  26.00-27.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  27.00-28.00  sec  1.19 MBytes  9.99 Mbits/sec  1041
[  5]  28.00-29.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
[  5]  29.00-30.00  sec  1.19 MBytes  9.99 Mbits/sec  1042
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-30.00  sec  35.8 MBytes  10.0 Mbits/sec  0.000 ms  0/31250 (0%)  sender
[  5]   0.00-30.10  sec  17.4 MBytes  4.84 Mbits/sec  0.181 ms  16076/31250 (51%)  receiver

iperf Done.
himasree@LAPTOP-41COT5AA:~/cnnp#
```

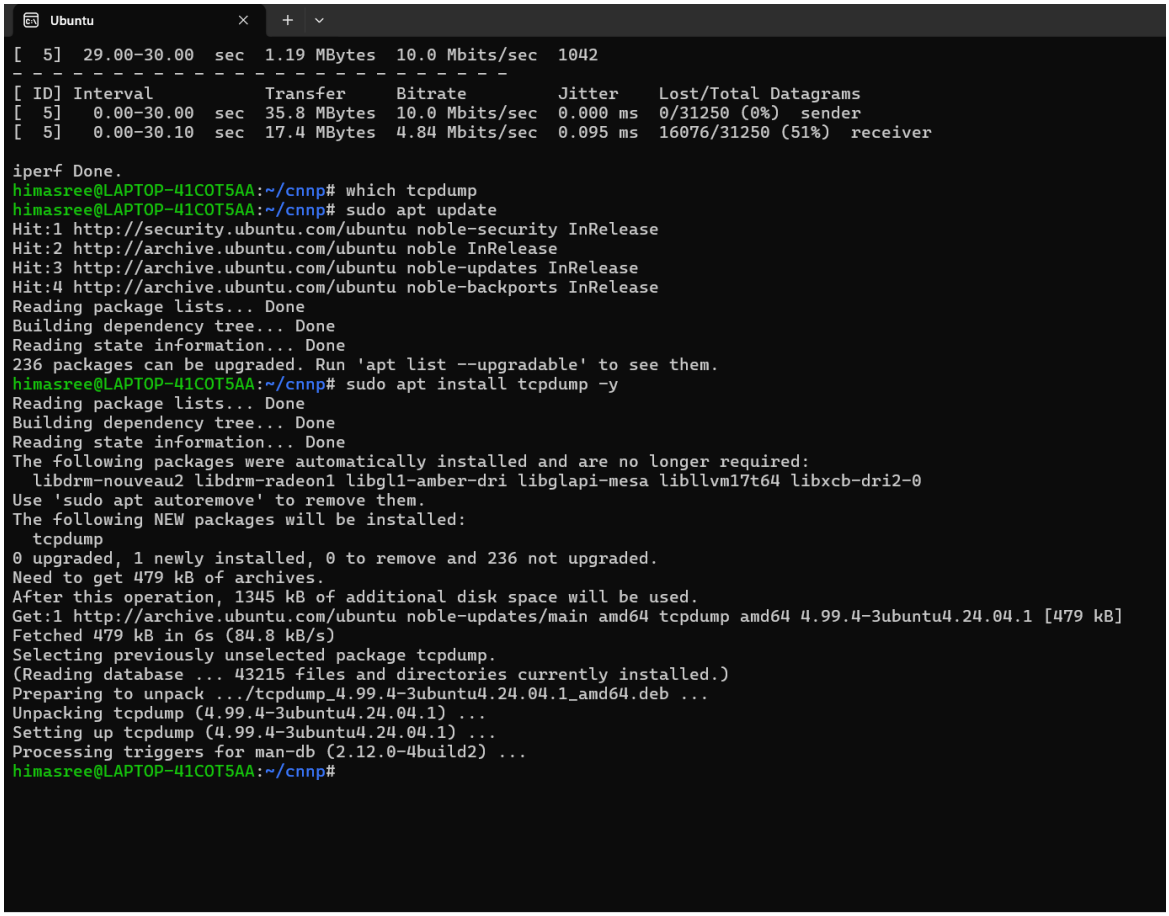
Receiver side (earlier run)



Receiver side (earlier run)

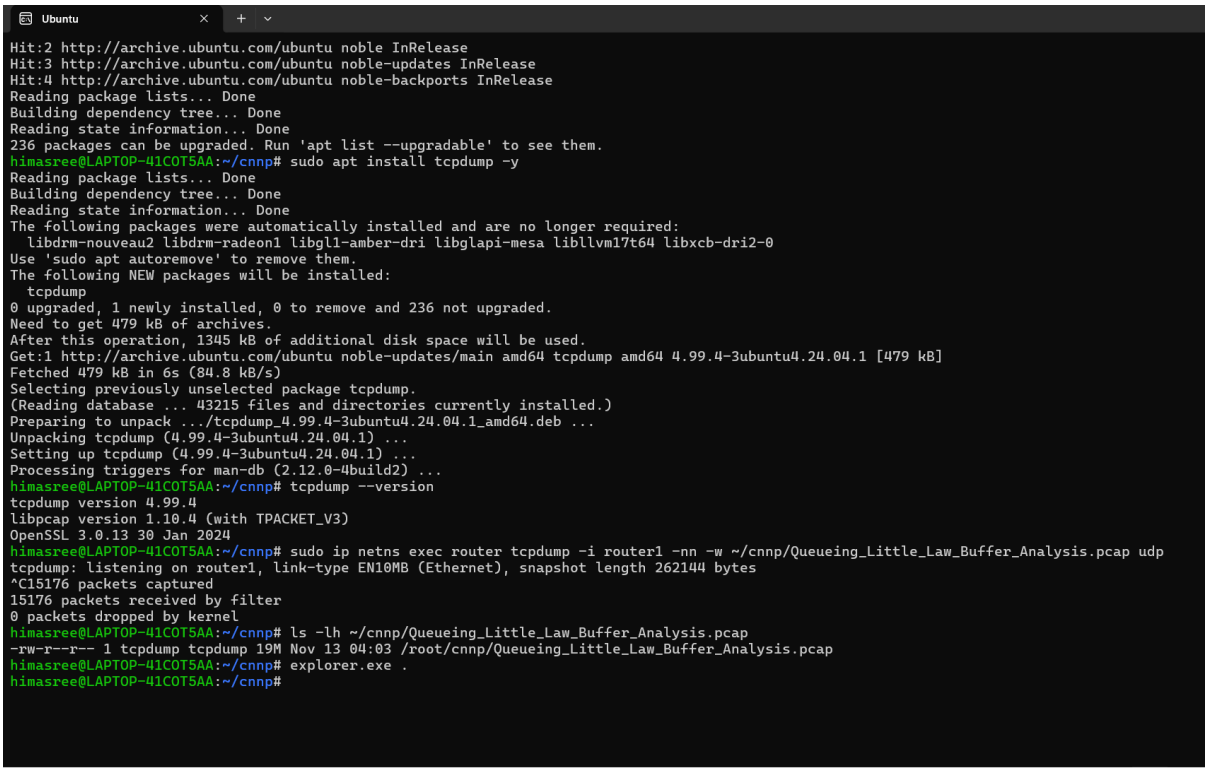


Step 6 – Capture packets using Wireshark



```
Ubuntu
[  5]  29.00-30.00  sec  1.19 MBytes  10.0 Mbits/sec  1042
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-30.00  sec  35.8 MBytes  10.0 Mbits/sec  0.000 ms  0/31250 (0%)  sender
[  5]   0.00-30.10  sec  17.4 MBytes  4.84 Mbits/sec  0.095 ms  16076/31250 (51%)  receiver

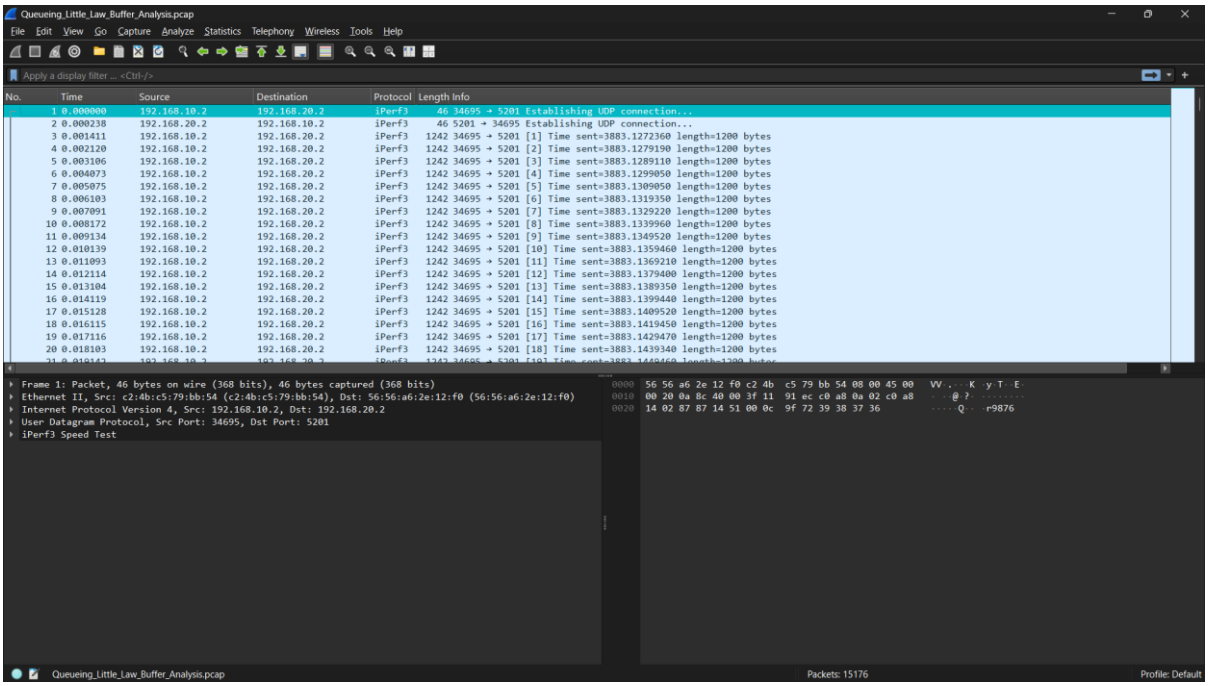
iperf Done.
himasree@LAPTOP-41COT5AA:~/cnnp# which tcpdump
himasree@LAPTOP-41COT5AA:~/cnnp# sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
236 packages can be upgraded. Run 'apt list --upgradable' to see them.
himasree@LAPTOP-41COT5AA:~/cnnp# sudo apt install tcpdump -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa libllvm17t64 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  tcpdump
0 upgraded, 1 newly installed, 0 to remove and 236 not upgraded.
Need to get 479 kB of archives.
After this operation, 1345 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 tcpdump amd64 4.99.4-3ubuntu4.24.04.1 [479 kB]
Fetched 479 kB in 6s (84.8 kB/s)
Selecting previously unselected package tcpdump.
(Reading database ... 43215 files and directories currently installed.)
Preparing to unpack .../tcpdump_4.99.4-3ubuntu4.24.04.1_amd64.deb ...
Unpacking tcpdump (4.99.4-3ubuntu4.24.04.1) ...
Setting up tcpdump (4.99.4-3ubuntu4.24.04.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
himasree@LAPTOP-41COT5AA:~/cnnp#
```

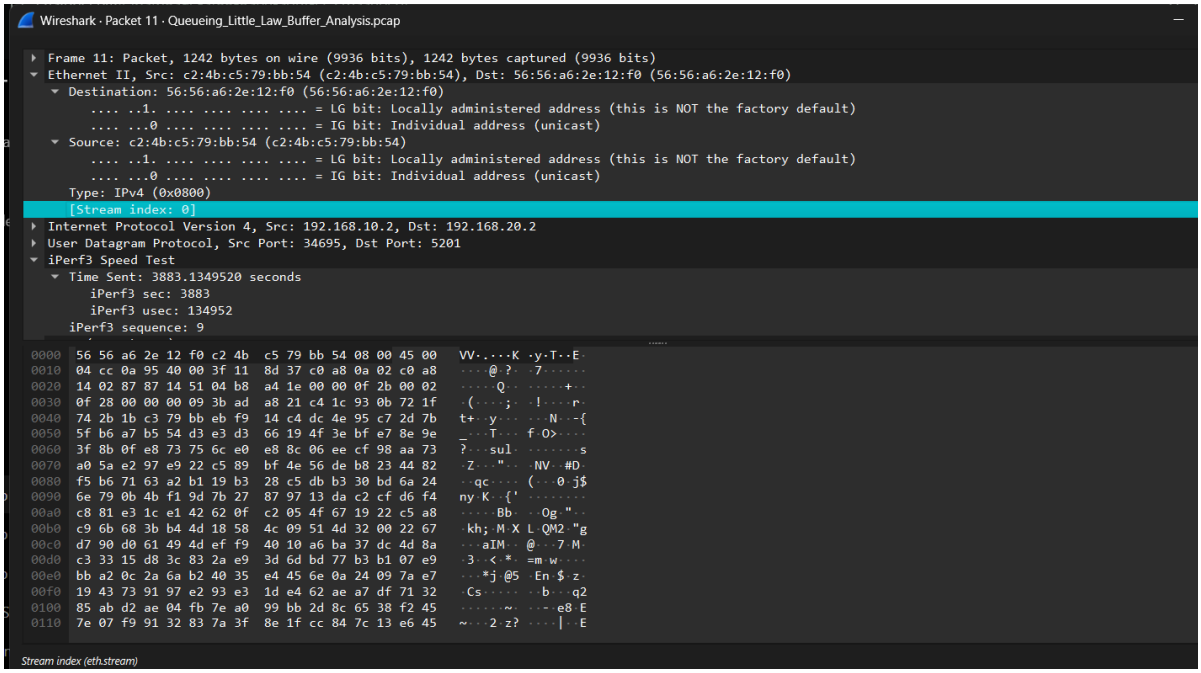


```
Ubuntu
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
236 packages can be upgraded. Run 'apt list --upgradable' to see them.
himasree@LAPTOP-41COT5AA:~/cnnp# sudo apt install tcpdump -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  libdrm-nouveau2 libdrm-radeon1 libgl1-amber-dri libglapi-mesa libllvm17t64 libxcb-dri2-0
Use 'sudo apt autoremove' to remove them.
The following NEW packages will be installed:
  tcpdump
0 upgraded, 1 newly installed, 0 to remove and 236 not upgraded.
Need to get 479 kB of archives.
After this operation, 1345 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 tcpdump amd64 4.99.4-3ubuntu4.24.04.1 [479 kB]
Fetched 479 kB in 6s (84.8 kB/s)
Selecting previously unselected package tcpdump.
(Reading database ... 43215 files and directories currently installed.)
Preparing to unpack .../tcpdump_4.99.4-3ubuntu4.24.04.1_amd64.deb ...
Unpacking tcpdump (4.99.4-3ubuntu4.24.04.1) ...
Setting up tcpdump (4.99.4-3ubuntu4.24.04.1) ...
Processing triggers for man-db (2.12.0-4build2) ...
himasree@LAPTOP-41COT5AA:~/cnnp# tcpdump --version
tcpdump version 4.99.4
libpcap version 1.10.4 (with TPACKET_V3)
OpenSSL 3.0.13 30 Jan 2024
himasree@LAPTOP-41COT5AA:~/cnnp# sudo ip netns exec router tcpdump -i router1 -nn -w ~/cnnp/Queueing_Little_Law_Buffer_Analysis.pcap udp
tcpdump: listening on router1, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C15176 packets captured
15176 packets received by filter
0 packets dropped by kernel
himasree@LAPTOP-41COT5AA:~/cnnp# ls -lh ~/cnnp/Queueing_Little_Law_Buffer_Analysis.pcap
-rw-r--r-- 1 tcpdump tcpdump 19M Nov 13 04:03 /root/cnnp/Queueing_Little_Law_Buffer_Analysis.pcap
himasree@LAPTOP-41COT5AA:~/cnnp# explorer.exe .
himasree@LAPTOP-41COT5AA:~/cnnp#
```

```
Ubuntu
15335 packets received by filter
0 packets dropped by kernel
himasree@LAPTOP-41COT5AA:~/cnnp# ip netns exec sender iperf3 -c 192.168.20.2 -u -b 1M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
[  5] local 192.168.10.2 port 41675 connected to 192.168.20.2 port 5201
[ ID] Interval           Transfer     Bitrate         Total Datagrams
[  5]   0.00-1.00   sec   123 KBytes  1.01 Mbits/sec  105
[  5]   1.00-2.00   sec   122 KBytes   998 Kbits/sec  104
[  5]   2.00-3.00   sec   122 KBytes   998 Kbits/sec  104
[  5]   3.00-4.00   sec   122 KBytes   998 Kbits/sec  104
[  5]   4.00-5.00   sec   122 KBytes   998 Kbits/sec  104
[  5]   5.00-6.00   sec   122 KBytes   998 Kbits/sec  104
[  5]   6.00-7.00   sec   123 KBytes  1.01 Mbits/sec  105
[  5]   7.00-8.00   sec   122 KBytes   998 Kbits/sec  104
[  5]   8.00-9.00   sec   122 KBytes   998 Kbits/sec  104
[  5]   9.00-10.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  10.00-11.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  11.00-12.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  12.00-13.00  sec   123 KBytes  1.01 Mbits/sec  105
[  5]  13.00-14.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  14.00-15.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  15.00-16.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  16.00-17.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  17.00-18.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  18.00-19.00  sec   123 KBytes  1.01 Mbits/sec  105
[  5]  19.00-20.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  20.00-21.00  sec   122 KBytes   999 Kbits/sec  104
[  5]  21.00-22.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  22.00-23.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  23.00-24.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  24.00-25.00  sec   123 KBytes  1.01 Mbits/sec  105
[  5]  25.00-26.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  26.00-27.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  27.00-28.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  28.00-29.00  sec   122 KBytes   998 Kbits/sec  104
[  5]  29.00-30.00  sec   122 KBytes   996 Kbits/sec  104
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bitrate         Jitter    Lost/Total Datagrams
[  5]   0.00-30.00  sec  3.58 MBytes  1000 Kbits/sec  0.000 ms  0/3125 (0%)  sender
[  5]   0.00-30.00  sec  3.58 MBytes  1000 Kbits/sec  0.025 ms  0/3125 (0%)  receiver

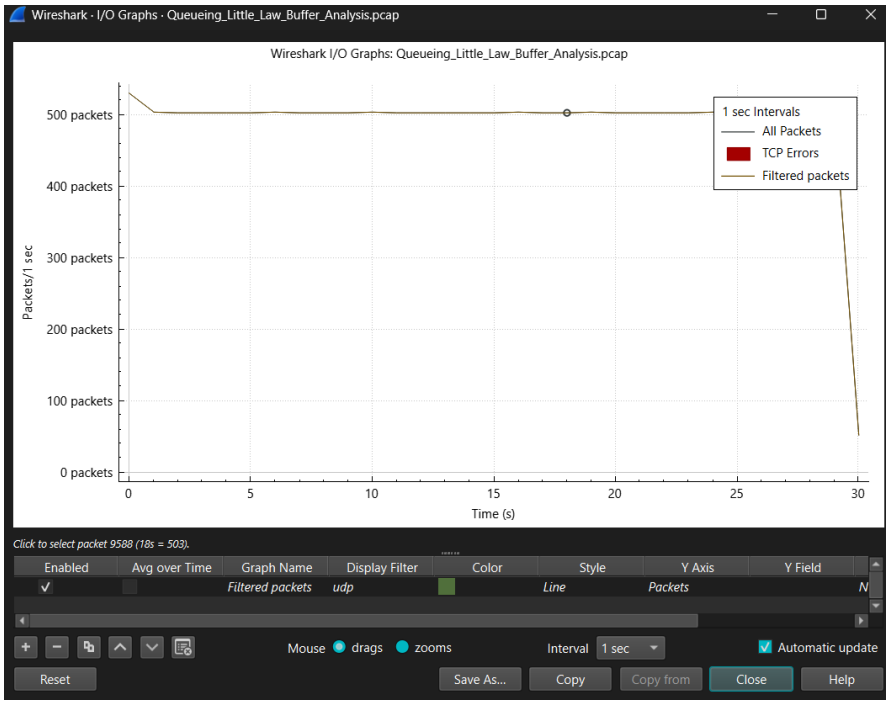
iperf Done.
himasree@LAPTOP-41COT5AA:~/cnnp# ip netns exec sender iperf3 -c 192.168.20.2 -u -b 1M -l 1200 -t 30 -i 1
Connecting to host 192.168.20.2, port 5201
```



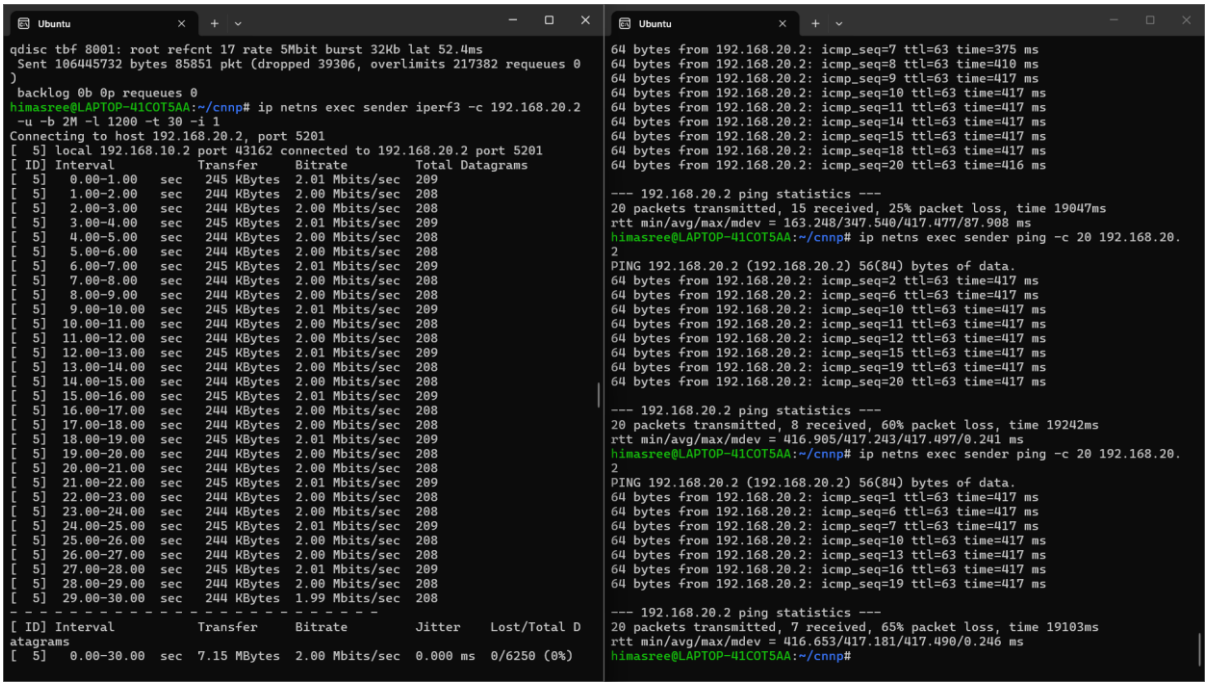


Step 7 – Analyze the queueing curve

Wireshark I/O Graph (filter: udp)



Baseline and loaded ping



Baseline RTT (ping -c 20, no traffic) = 0.106 ms. Delay above baseline = loaded average RTT − baseline RTT.

$$5 \text{ Mbps: } 347.540 - 0.106 = 347.434 \text{ ms}$$

$$7 \text{ Mbps: } 417.243 - 0.106 = 417.137 \text{ ms}$$

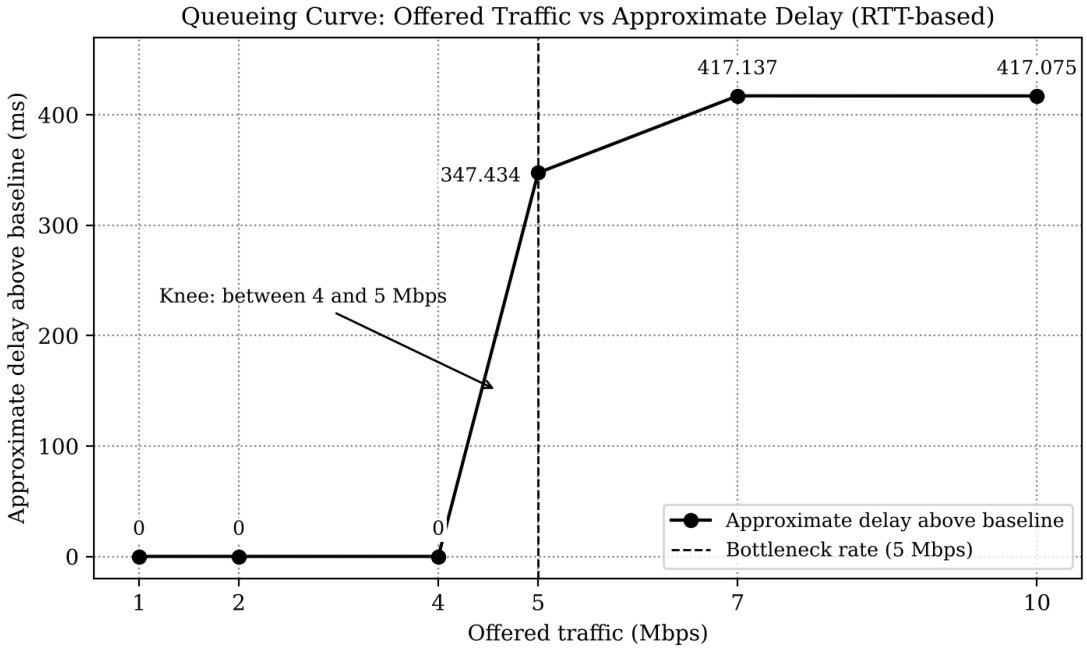
$$10 \text{ Mbps: } 417.181 - 0.106 = 417.075 \text{ ms}$$

For 1, 2 and 4 Mbps the loaded RTT (0.058, 0.054, 0.049 ms) is below the baseline, so the delay is taken as approximately 0 ms.

Table 2: Loaded RTT and delay above baseline

Offered traffic (Mbps)	Loaded average RTT (ms)	Ping loss (%)	Delay above baseline (ms)
1	0.058	0	≈ 0
2	0.054	0	≈ 0
4	0.049	0	≈ 0
5	347.540	25	347.434
7	417.243	60	417.137
10	417.181	65	417.075

Queueing curve



The knee of the curve is between 4 and 5 Mbps, where the delay rises sharply.

Step 8 – Buffer analysis

tc -s qdisc show dev router1

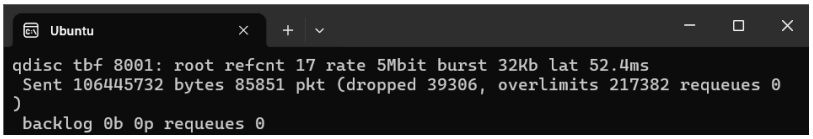


Table 3: Buffer analysis (10 Mbps offered, 5 Mbps bottleneck)

Buffer limit	Maximum observed backlog	Packet drops	Ping RTT	Receiver throughput (Mbps)	Packet loss (%)	Jitter (ms)
16 KB	Not available	Not available	Not available	4.84	52	0.126
64 KB	Not available	Not available	Not available	4.84	51	0.088
256 KB	Not available	Not available	Not available	4.84	51	0.273

Step 9 – Experimentally validate Little's Law

A. Packet arrival rate (λ)

10 Mbps test: total datagrams = 31,250, lost = 15,917, so received = 31,250 − 15,917 = 15,333. Receiver duration = 30.42 s.

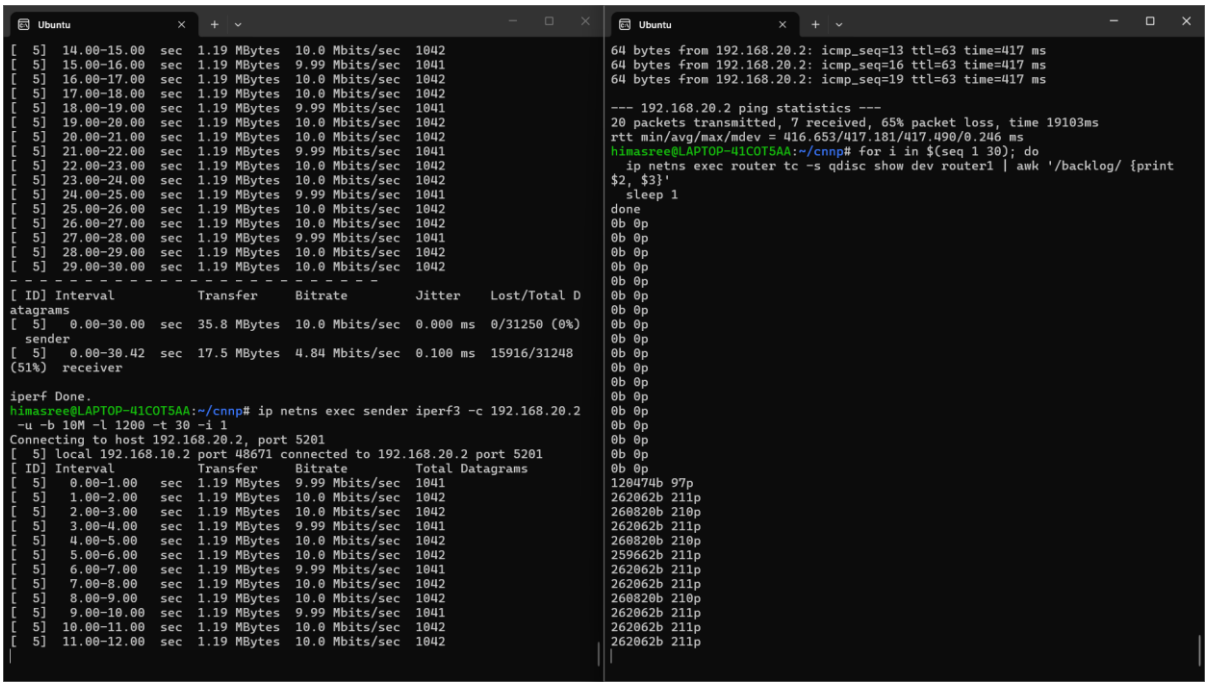
$$\lambda = 15333 / 30.42 \approx 504 \text{ packets/s}$$

B. Queueing delay (W) from ping

$$W = 417.181 - 0.106 = 417.075 \text{ ms} = 0.417 \text{ s}$$

C. Average queue length (L)

Backlog sampling (30 samples)



Samples 1–18 = 0 packets; samples 19–30 = 97, 211, 210, 211, 210, 211, 211, 211, 210, 211, 211, 211 packets.

$$\text{Sum} = 2415 \text{ packets}$$

$$\text{Measured average } L = 2415 / 30 = 80.5 \text{ packets}$$

D. Little's Law

$$L = \lambda \times W = 504 \times 0.417 \approx 210 \text{ packets}$$

$$W = L / \lambda = 80.5 / 504 = 0.160 \text{ s} \approx 160 \text{ ms}$$

The predicted L (about 210 packets) is close to the maximum sampled backlog (211 packets) but not to the measured average (80.5 packets), because the queue was empty for the first 18 samples. The Little's Law delay (160 ms) is lower than the ping-based delay (417 ms) for the same reason; the two do not match exactly. Note: the backlog was sampled while tc showed lat 367ms (about a 256 KB limit); the configuration was re-applied afterwards and verified as lat 52.4ms (64 KB).

Student observation and record

Table 4: Observation table – Queueing curve

Test	Offered traffic	Received throughput	Avg. delay	Packet loss	Ping loss
1	1 Mbps	1.00 Mbps	≈ 0 ms	0%	0%
2	2 Mbps	2.00 Mbps	≈ 0 ms	0%	0%
3	4 Mbps	4.00 Mbps	≈ 0 ms	0%	0%
4	5 Mbps	4.84 Mbps	347.434 ms	1.9%	25%
5	7 Mbps	4.84 Mbps	417.137 ms	30%	60%
6	10 Mbps	4.84 Mbps	417.075 ms	51%	65%

Table 5: Observation table – Buffer analysis

Test	Buffer limit	Maximum observed backlog	Packet drops	Ping RTT	Throughput / loss
1	16 KB	Not available	Not available	Not available	4.84 Mbps / 52%
2	64 KB	Not available	Not available	Not available	4.84 Mbps / 51%
3	256 KB	Not available	Not available	Not available	4.84 Mbps / 51%

Table 6: Observation table – Little's Law

Parameter	Measured value
Total packets observed	15,333 packets
Observation time (seconds)	30.42 s
Arrival rate λ (packets/sec)	≈ 504
Baseline RTT (ms)	0.106
Loaded RTT (ms)	417.181
Estimated queueing delay W (seconds)	0.417
Calculated $L = \lambda \times W$	≈ 210 packets
Measured average queue length	80.5 packets